

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc.
Project Name: Hilo WWTP Phase 1
Submittal Title: Bins
TO: County
From: Nan Inc

Project No.: WW-4705R
Submittal Number: 12330 - 002.0

☐ For Information Only
☐ Designer of Record
☐ Government Approval

Specification No. and Subject of Submittal / Equipment Supplier	
Spec: 12330	Paragraph: 1.02, 2.01, 2.02
Authored By: Park USA	Date Submitted: 06/09/2025

Submittal Certification	
Check Either (A) or (B):	
☒ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: Jyun-Cheng Jhuo Digitally signed by Jyun-Cheng Jhuo DN: cn=Jyun-Cheng Jhuo, o=Nan Inc., cn=Jyun-Cheng Jhuo Date: 2025.06.09 16:11:53 -0500	
Printed Name and Title: Jyun-Cheng Jhuo	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 6/16/2025
SPECIFICATION SECTION # 12330
SPECIFICATION Bins
PARAGRAPH 1.02, 2.01, 2.02
DRAWING N/a
SUBCONTRACTOR N/a
SUPPLIER N/a
MANUFACTURER Park Process

CERTIFIED BY CQCM or Designee :

Station 102

SECTION 12330

BINS

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes: Equipment for disposal of screenings and grit generated at the Septage Receiving Station (Area 01) and Headworks Facility (Area 02).

1.02 SUBMITTALS

- ✓ A. Submit: As specified in Section 01330 - Submittal Procedures and Section 15050 - Common Work Results for Mechanical Equipment.
- ✓ B. Product data: Provide at a minimum dimensions, configurations, construction details, joint details, materials of construction, color selection chart, gate, hinges, gasket material, and ball valve.
- ✓ C. Shop drawings. Submit in accordance with Section 01330 - Submittal Procedures and Section 15050 - Common Work Results for Mechanical Equipment.
 - 1. Provide letter confirming dewatering roll-off bins are compatible with a 26-foot ST6 Shorty-style roll-off trailer.
- D. Warranty conforming to Section 01783 - Warranties and Bonds.
- E. Operation and Maintenance Manual. Conform to Section 01782 - Operation and Maintenance Manuals.
- F. Manufacturer's installation instructions: Indicate special installation requirements.

1.03 QUALITY ASSURANCE

- A. Manufacturer qualifications: Company specializing in manufacturing the products specified in this Section with minimum 5 years of experience.

1.04 DELIVERY, STORAGE, AND HANDLING

- A. Delivery:
 - 1. Deliver vector truck dewatering bin within the first 365 days of Notice to Proceed.
 - 2. Screenings and grit bin, septage bin, and screenings bypass bin:
 - a. Do not deliver to site until destination space is ready to receive it.
 - b. Deliver prior to start-up and commissioning of the septage and headworks facilities.
 - 3. Deliver bins suitably wrapped or packaged to protect against damage.
 - a. Do not remove protective coverings until time of installation.

- B. Accept bins on site.
- C. Inspect on arrival for damage.
- D. Coordinate size of access and route to place of installation.

1.05 FIELD MEASUREMENTS

- A. Verify that field measurements are as indicated on shop drawings.

1.06 COORDINATION

- A. Coordinate bin installation with size, location, and installation of other equipment as indicated on the Drawings.

PART 2 PRODUCTS

2.01 DESIGN REQUIREMENTS

- A. Roll-off dumpster:
 - ✓ 1. ✓ Manufacturer: The following or approved equal:
 - ✓ a. ParkUSA, GritCat Model GCRO-17-T-PCS.
 - ✓ 2. Construction:
 - a. Dewatering container shall be constructed of A36 carbon steel plate with 1/4-inch plate floor and 3/16-inch plate walls.
 - b. Round bottom design.
 - c. Compatible with 26-foot ST6 Shorty-style roll-off trailer.
 - ✓ 3. Drains:
 - a. Number: Minimum 4 each container.
 - b. Size: 3 inch.
 - c. Provide 3-inch ball valve with a stainless-steel hose camlock connection with a stainless steel camlock cap and chain.
 - ✓ 4. Hook attachments:
 - a. Number: 2 at each end of container to allow pulling from either end.
 - b. Type: Steel D-ring or similar attached to container frame.
 - c. Hook plate thickness: 1.25 inch A36 steel.
 - ✓ 5. Gates:
 - a. Latch: adjustable and located on driver's side.
 - b. Hinges: Heavy duty with grease fittings.
 - c. End door seal: Liquid tight with replaceable gasket.
 - ✓ 6. Welds:
 - a. Interior: Continuous.
 - b. Exterior: Skip welds with caulking between welds. Caulk shall be 25-year caulking applied before coating.
 - ✓ 7. Coating:
 - a. Coat in accordance with Section 09960 - High-Performance Coatings.
 - b. Provide white label lettering as direct by the Engineer and Owner.

- ✓ 8. Filter panels:
 - a. Provide removable internal filter panels (walls and floor) compatible with the specified bins.
 - 1) Model: Park Process PPFM-26-053P or approved equal.
 - b. Fiber material: Polyester.
 - c. Micron retention: 260 microns.
- ✓ 9. Roller track:
 - a. Bin provided shall be suitable for roller tracks as shown on the Drawings.
- ✓ 10. Tarp:
 - a. A 22-ounce vinyl coated nylon tarp shall be installed on the top of the dewatering container. This tarp shall be a side-to-side rolling tarp and shall come complete with tarp bows, tarp rolling hardware, ratchets and straps, bungee tie-down cords, and aluminum tarp stops.
- ✓ 11. As indicated in the following schedule:

Container	Screenings and Grit Bin	Vactor Truck Dewatering Bin
Number of Containers	2	1
Type	Dewatering - Roll-Off	Dewatering - Roll-Off
Storage Volume (cubic yards)	20	20
Container Dimensions	Length: 23-feet-0 inches Width: 8-feet-0 inches Height: 4-feet-0 inches	Length: 23-feet-0 inches Width: 8-feet-0 inches Height: 4-feet-0 inches
Center to Center of Rollers	7-feet-0 inches	7-feet-0 inches
Roller Dimensions	Width: 0-feet-8 inches Diameter: 0-feet-8 inches	Width: 0-feet-8 inches Diameter: 0-feet-8 inches
Nose Roller Dimensions	Width: 4 to 6 inches. Diameter: 4 to 6 inches. Axle: 1.5 inches with grease fittings.	Width: 4 to 6 inches. Diameter: 4 to 6 inches. Axle: 1.5 inches with grease fittings.
Filter Panels	Not required	Required

- B. Self-dumping dumpster:
 - 1. Manufacturer: The following or approved equal:
 - ✓ a. ParkUSA, GritCat Model DH.
 - ✓ 2. Construction:
 - a. Dewatering container shall be constructed of 7 gauge steel, with structural angle frame and structural base.
 - b. Base shall be equipped with a one piece flame cut steel track on each side and rocker stops for smooth operation and alignment.
 - c. The container shall meet ANSI safety specifications and dimensional standards. While container is in use, it shall be locked in place by a safety release latch.

- d. When emptying the load in the container, the unit is tilted back, the safety latch is released, the unit is tilted forward and the container rocks forward to dump its contents.
- e. Tilting the unit back after emptying shall assist in returning the hopper to its latched position.
- f. The container shall be designed to be handled by a standard forklift (making sure the weight of the container and load do not exceed the capacity of the forklift).
- ✓ 3. Drains:
 - a. Number: Minimum 2 each container.
 - b. Size: 2-inch.
 - c. Female threads equipped with male stainless steel camlock fittings and watertight camlock caps.
- ✓ 4. Welds:
 - a. Interior: Continuous.
 - b. Exterior: Skip welds with caulking between welds. Caulk shall be 25-year caulking applied before coating.
- ✓ 5. Coating:
 - a. Coat in accordance with Section 09960 - High-Performance Coatings.
 - b. Provide white label lettering as direct by the Engineer and Owner.
- ✓ 6. Filter panels:
 - a. Provide removable internal filter panels (walls and floor) compatible with the specified bins.
 - b. Fiber Material: Polyester.
 - c. Micron retention: 260 microns.
- ✓ 7. As indicated in the following schedule:

Container	Septage Bin
Number of Containers	1
Type	Dewatering - Self-Dumping
Storage Volume (cubic yards)	2
Container Dimensions	Length: 6-feet-2 inches Width: 4-feet-4 inches Height: 4-feet-0 inches
Roller Dimensions	Four 6-inch heavy duty caster wheels with stainless steel components
Filter Panel	Required

- ✓ C. Dumpster:
1. As indicated in the following schedule:

Container	Screenings Bypass Bin
Number of Containers	2
Type	Dewatering - Self-Dumping
Storage Volume (cubic yards)	2
Container Dimensions	Length: 4-feet-0-inches minimum/ 5-feet-4-inches maximum Width: 3-feet-0-inches maximum Height without rollers: +/- 4-feet-0-inches
Center to Center of Rollers	NA
Roller Dimensions	Four 6-inch heavy duty caster wheels with stainless steel components
Drains	Provide 1.5-inch replaceable shutoff valve (ball valve).
Nose Roller Dimensions	NA
Hook Attachments	NA
Gates	NA
Welds	Interior: Continuous. Exterior: Skip welds with caulking between welds. Caulk shall be 25-year caulking applied before coating.
Coating	Coat in accordance with Section 09960 - High-Performance Coatings.
Roller Track	NA
Additional Comments:	Provide bin with frame so it can be lifted and moved with a forklift.

2.02 FABRICATION

- ✓ A. Fabricate in accordance with Section 05500 - Metal Fabrications and the design requirements.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify conditions are suitable to receive the work of this Section.

END OF SECTION



Submittal

Customer: Nan Inc / Promark

Contact Name: Freddie Leonar

Location: Hilo WWTP

Job Number: N/A

Quote Number: 25-70779

ParkProcess Equipment:

- 17 yd GritCat Roll-Off
- 2 yd GritCat Dump Hopper



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- **SECTION 7** - Drawings
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SECTION - 1

Equipment Specification



Vacuum Truck Waste Dewatering Containers

Container Construction

Dewatering container shall be a Park Process GritCat Roll-Off- Model GCRO- 17-T-PCS as manufactured by Park Process of Houston, TX

(855-511-PARK). Alternates or substitutions shall require approval.

Dimensions shall not exceed 23'L x 8'6" W x 4' H nor shall it weigh over 8,700 lbs. Internal volume shall yield a minimum of 12 cubic yards of waste cake holding capacity.

Dewatering container shall be constructed of A-36 carbon steel plate with 1/4" plate floor and 3/16" plate walls. The unit shall be of a round

bottom design with a 12" radius on the walls where they join the floor plate. Floor cross members shall be made from C3 4.1 lb. channel and shall be

spaced on 12" centers.

Dewatering container shall have two 3" threaded drainage ports on each lower side at opposing ends. Each port shall have a 3" stainless steel hose camlock connection with a stainless steel camlock cap and chain, aluminum or other material will not be allowed. These ports shall be coped into the bottom radius of the container effectively draining water from the installed interior filter system. The drainage system shall drain effluent water from any one of or all 4 of the drainage ports.

Container long sills (rails) shall be constructed of 6" x 2" x 1/4" structural tubing with inside spacing of 36.5".

1.6 Triangular gussets shall be welded to the outside of both rails and to each of the 12" spaced floor cross members for extra strength.



The dewatering container shall be equipped with four (4) wheels, 8" diameter x 8" wide diagonally braced constructed carbon steel. Tracking

wheels shall be 4" diameter x 6" wide and made of carbon steel. All wheel axles shall have grease fittings.

Wall gussets shall be made of ¼" plate.

Door sealing face will be totally watertight and have a "T" style EPDM gasket installed around the perimeter of the door face. Door frame shall be made of 4" x 3" x 3/16" structural tubing with 3 steel hinges located on the side. Door sealing shall have 1" ratchets and chain binders, one on each side and two in the middle of the door. Steel 3/8" chain shall be used with a floating ring for easily aligning the chain and ratchet. Hinges are constructed of ½" wall steel tubing, 1" diameter hot rolled bar pins

with welded retainer washers, and 1" x 3 ½" x 9" carbon steel hinge plates and shall have grease fittings. Door sheet will be made from 3/16" plate.

All weld seams inside and outside of the container shall be seal welded for better corrosion protection. Skip welding on the exterior weld joints and using calk prior to painting is not allowed.

Internal filter support system shall be comprised of carbon steel ¼" angles and cold formed channels welded to the inside of the container with the removable filter support panels

bolted to the welded-in structure. Wall support panels shall be made hot dip galvanized #9 flattened expanded metal with galvanized carbon steel flat bar frames welded around the perimeter of

each panel, Wall support panels shall run the length of the container on both long walls and across the front of the container. The floor support panels

shall be made of 3/16" thick carbon steel perforated plate. The floor perforated plate panels shall be hot

dip galvanized. The wall support panels and floor panels shall be bolted into place using stainless steel fasteners consisting of set screws, flat washers, lock washers and acorn nuts.



The filter media shall be Park Process PPFM-26-053P and shall have the following characteristics:

Color: Blue

Fiber: Polyester

Count: 68 x 20 per inch

Weave: 6 x 2 modified satin

Warp Diameter: 500 microns

Weft Diameter: 800 microns

Weight: 42.5 oz. per sq. yd.

Tensile strength: 1560 lbs.per inch

Air Permeability: 385 mm³/mm²/sec- 325-355 CFM

Thickness: .085"

Micron opening: 0 x 325

Micron retention: 260

Open area: 19%

A 22 ounce vinyl coated nylon tarp shall be installed on the top of the dewatering container. This tarp shall be a side to side rolling tarp and shall come complete with tarp bows, tarp rolling

hardware, ratchets and straps, bungee tie down cords, and aluminum tarp stops.

Dewatering Container Surface Prep and Coating

Exterior - All steel metal surfaces shall be sandblasted to near white metal with a SP-10 blast, with 2-3 mils of anchor profile, primed using Carboguard 890 white at 4-6 mils dry, and top

coated with Carbothane Park Process blue at 2-3 mils. Paint shall be Carboline brand, no substitutions allowed.



Interior - All steel metal surfaces except the filter support panels and tie down bars shall be sandblasted to near white metal with a SP-10 blast, with 2 to 3 mils of anchor profile, then coated with two coats of Carboguard Park Process blue 4 to 6 dry mils per coat to a total of 10 to 12 dry mils.

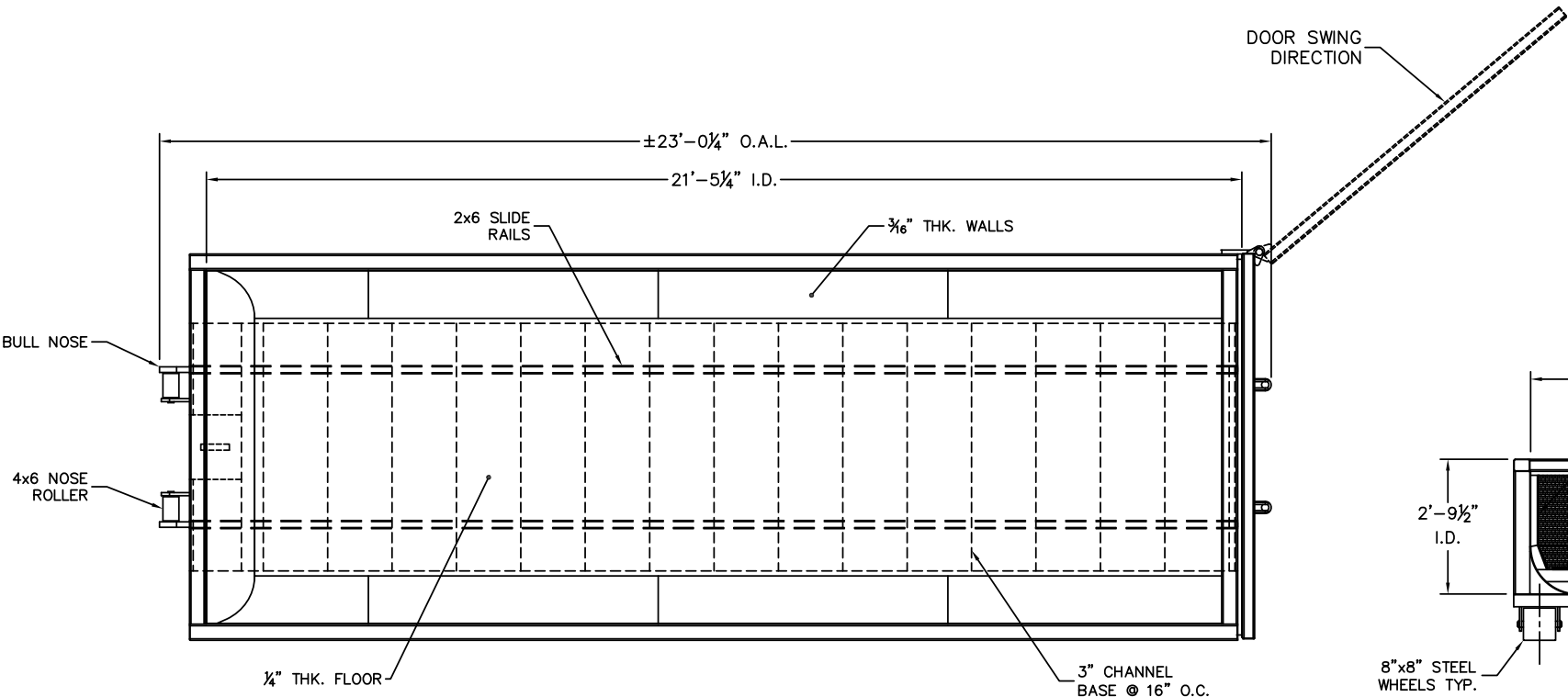
Paint shall be Carboline, no substitutions allowed.

The expanded metal perforated plate support panels shall be hot dip galvanized

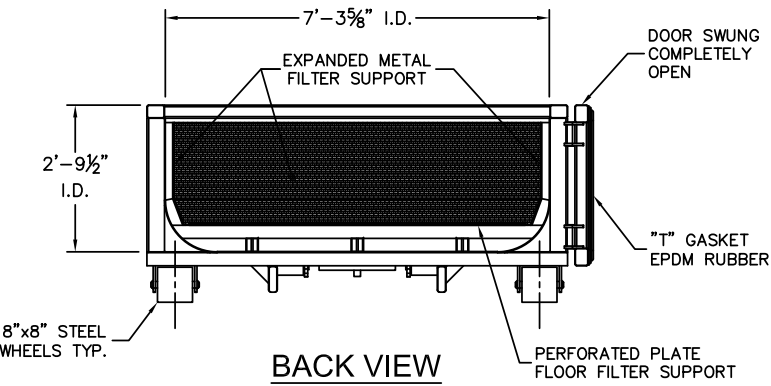
SECTION - 2

Drawings

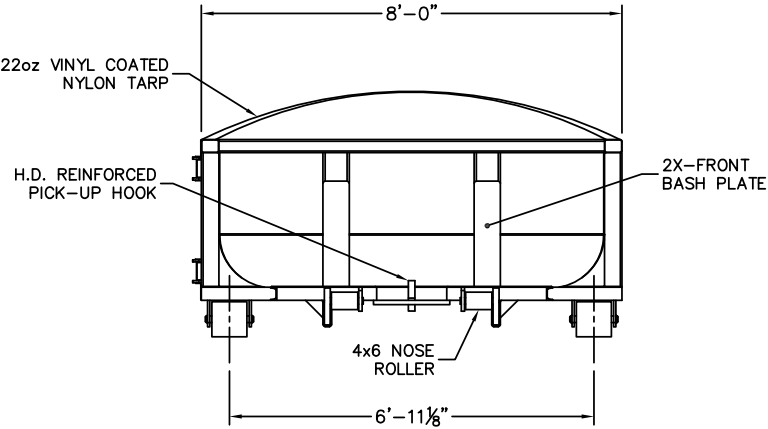
8,500 lbs APPROX



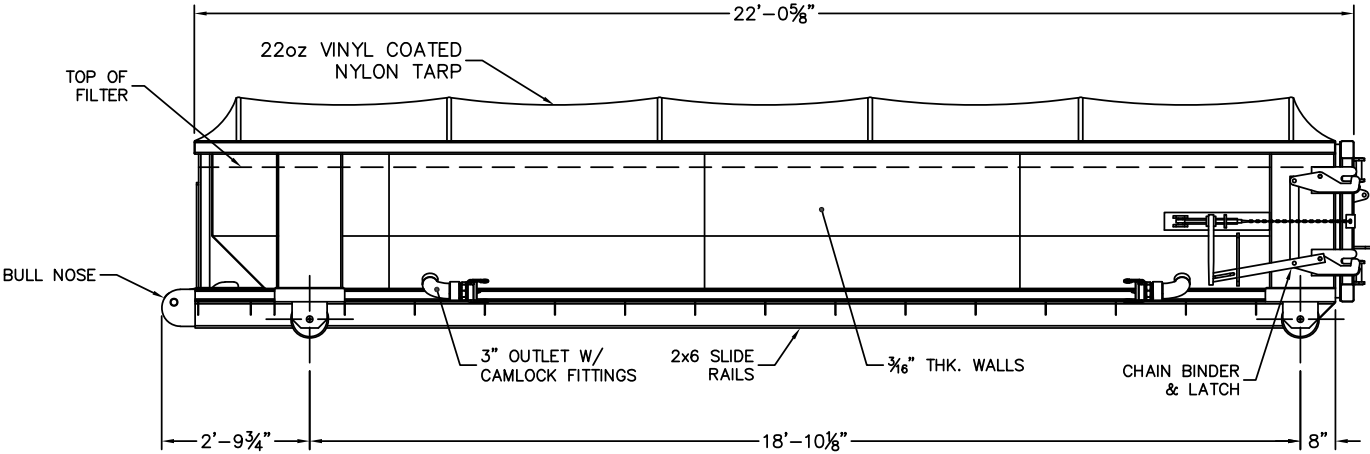
TOP VIEW
FILTERS & TARP NOT SHOWN FOR CLARITY



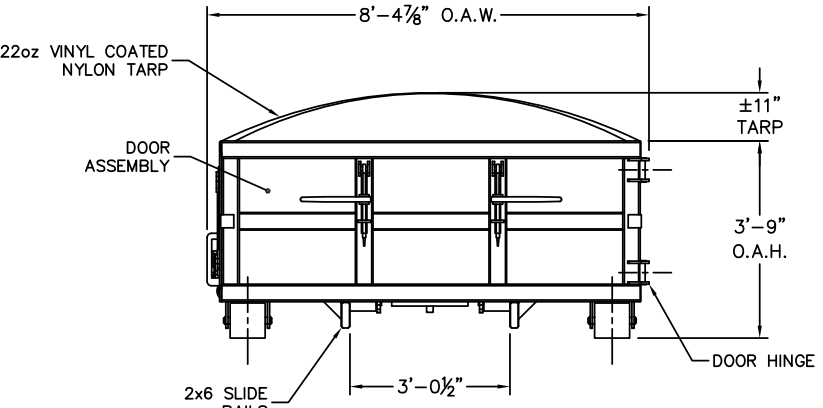
BACK VIEW
DOOR OPEN
TARP NOT SHOWN FOR CLARITY



FRONT VIEW



SIDE VIEW



BACK VIEW
DOOR CLOSED

CUSTOMER APPROVAL:
SIGNED APPROVAL IS REQUIRED BEFORE
BEGINNING CONSTRUCTION/FABRICATION.

PRINT NAME

SIGNATURE

DATE (MM/DD/YYYY)

NOTES:

1. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED OTHERWISE.
2. CAPACITY: 15 CUBIC YARDS
3. THIS MODEL IS EQUIPPED WITH FILTERS ON THE FLOOR, SIDES, AND FRONT WALLS.
4. TRUCK COMPATIBILITY: HOIST & CABLE

THIS SHEET IS THE PROPERTY OF PARK PROCESS, HOUSTON, TEXAS, AND MAY CONTAIN PROPRIETARY INFORMATION. IT IS CIRCULATED UNDER THE EXPRESS CONDITION THAT IT IS NOT TO BE USED IN ANY MANNER DIRECTLY OR INDIRECTLY DETRIMENTAL TO PARK PROCESS.

WEIGHTS, DIMENSIONS AND DESIGN ARE SUBJECT TO CHANGE. CONTACT PARK PROCESS FOR THE MOST UP TO DATE EQUIPMENT DRAWINGS AND INFORMATION.



17 YARD GRITCAT ROLL-OFF
GENERAL ARRANGEMENT

PM TJ	DRN CH	CHK FG	DWG. NO. PP-GCRO-17-GA-01	REV. A
DATE	2/27/25			

SECTION - 3

Cut Sheets

SECTION - 3.A

Cam Lock Fittings



P-T Coupling Co., Inc.
1414 E. Willow
Enid,
Phone:580-237-4060
Website: <http://www.ptcoupling.com>

Item # 1000630, F-Adapter (Adapter x Male NPT Thread)
List Price QUOTE

PT maintains inventories at its 16 branch warehouse locations.

HBS cam arms are made of 300 series Stainless Steel.

HB cam arms are made of Brass.

All Stainless Steel castings are 316 and all components are 300 series Stainless Steel.

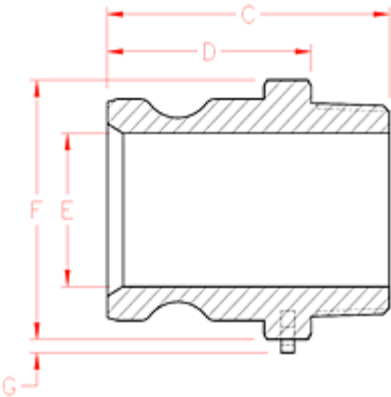
Please call for special quote on complete cam arm assembly and components in 316 Stainless Steel.

8, 10 and 12 inch couplers are 4-cam designs with the STA-LOK® positive locking feature.

Couplings not interchangeable in the industry are sizes 8" through 12".

Made in the U.S.A.

Note: Prices subject to change without notice.



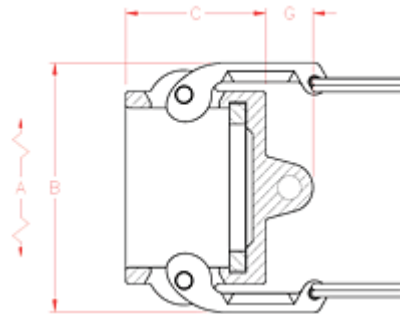
Specifications

Part Name	30F
Type	Adapter x Male NPT Thread
Material	Aluminum
Size	3 in.
Adapter Size	3 in.
Thread (Male NPT)	3 in.
Overall Length (C)	4.37 in. 111 mm
Exposed Length (D)	2.87 in. 73 mm
Inside Diameter (E)	2.80 in. 71 mm

Max. Across Corners (F)	4.07 in. 103 mm
Chain Lug Extension (G)	0.44 in. 11 mm
Engagement Length	1.84 in.

V-Coupler (Dust Cap)

- PT maintains inventories at its 16 branch warehouse locations
- HBS cam arms are made of 300 series Stainless Steel
- HB cam arms are made of Brass
- All Stainless Steel castings are 316 and all components are 300 series Stainless Steel
- 8,10 and 12 inch couplers are 4-cam designs with the STA-LOK® positive locking feature
- Couplings not interchangeable in the industry are sizes 8" through 12"

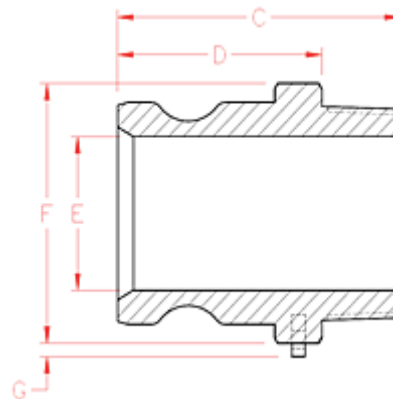


Specifications

Part Name	30V
Type	Dust Cap
Material	Aluminum/HB
Size	3 in.
Coupler Size	3 in.
O. D. with Cam Arms Extended (A)	9.56 in. 243 mm
Outside Diameter (B)	5.46 in. 139 mm
Overall Length (C)	2.27 in. 58 mm
Chain Lug Extension (G)	1.03 in. 26 mm
Engagement Length	1.84 in.

F-Adapter (Adapter x Male NPT Thread)

- PT maintains inventories at its 16 branch warehouse locations.
- HBS cam arms are made of 300 series Stainless Steel.
- HB cam arms are made of Brass.
- All Stainless Steel castings are 316 and all components are 300 series Stainless Steel.
- 8, 10 and 12 inch couplers are 4-cam designs with the STA-LOK® positive locking feature.
- Couplings not interchangeable in the industry are sizes 8" through 12".



Specifications

Part Name	30F
Type	Adapter x Male NPT Thread
Material	Aluminum
Size	3 in.
Adapter Size	3 in.
Thread (Male NPT)	3 in.
Overall Length (C)	4.37 in. 111 mm
Exposed Length (D)	2.87 in. 73 mm
Inside Diameter (E)	2.80 in. 71 mm
Max. Across Corners (F)	4.07 in. 103 mm
Chain Lug Extension (G)	0.44 in. 11 mm
Engagement Length	1.84 in.

SECTION - 3.B

Door Chain

NANTONG HEXING CHAIN CO.,LTD.

ADD: XINDIAN INDUSTRIAL ZONE,RUDONG COUNTY, NANTONG CITY, JIANGSU PROVINCE, CHINA

TO:ELITE SALES,INC,

PO Number:801280/801298/801313

LC Number: 237

PHYSICAL AND CHEMICAL MILL TEST CERTIFICATES

PHYSICAL AND MECHANICAL MILL TEST CERTIFICATES										CHEMICAL COMPOSITION							
DESCRIPTION OF GOODS	PCS	TOTAL LENGTH	SIZE			TENSILE TEST				C	Si	Mn	P	S	Cr	Ni	Mo
			D mm	L mm	b mm	BREAKING LOAD(LBS)	PROOF LOAD(LBS)	WORKING LOAD(LBS)	ELONGATIO %								
G30 PROOF COIL CHAIN EG 801280																	
3/16"X250FT	70	17500FT	5.5	24.38	7.7	3200	1600	800	15	0.13	0.14	0.38	0.022	0.035			
5/16"X550FT	20	11000FT	8	32.8	11.2	7600	3800	1900	15	0.13	0.14	0.38	0.022	0.035			
G30 PROOF COIL CHAIN HG 801280																	
3/16"X1500FT	10	15000FT	5.5	24.8	7.7	3200	1600	800	15	0.13	0.14	0.38	0.022	0.035			
7/8"X80FT	10	800FT	22	77	27.5	51200	25600	12800	15	0.13	0.14	0.38	0.022	0.035			
G70 BINDING CHAIN YZP 801280																	
5/16"X20FT	50	1000FT	8.7	32.8	11.2	18800	9400	4700	15	0.21	0.32	1.55	0.017	0.001			
G80 ALLOY CHAIN BLACK 801280																	
1/2"X200FT	25	5000FT	13	41.5	18.5	48000	24000	12000	15	0.21	0.32	1.55	0.017	0.001			
G70 TRANSPORT CHAIN YZP 801280																	
1/4"X800FT	25	20000FT	7	31.5	9.8	12600	6300	3150	15	0.21	0.32	1.55	0.017	0.001			
STAINLESS STEEL CHAIN 316 801280																	
1/8"X1500FT	5	7500FT	4	23.9	6.4	2400	1200	600	15	0.034	0.32	0.7	0.034	0.001	18.2	8.05	
G80 ALLOY CHAIN BLACK 801298																	
1/2"X200FT	50	10000FT	13	41.5	18.5	48000	24000	12000	15	0.21	0.32	1.55	0.017	0.001			
G30 PROOF COIL LONG CHAIN HG 801313																	
1/2"X200FT	14	2800FT	13	88.9	20.27	18000	9000	4500	15	0.13	0.14	0.38	0.022	0.035			
1"X100FT	5	500FT	25	104.78	34.93	71600	35800	17900	15	0.13	0.14	0.38	0.022	0.035			
5/8"X100FT	7	700FT	16	95.25	25.4	27600	13800	6900	15	0.13	0.14	0.38	0.022	0.035			
G43 BOOMER CHAIN SC 801313																	
1/2"X20FT	100	2000FT	13	45.5	18.2	27600	13900	9200	15	0.21	0.32	1.55	0.017	0.001			
G30 PROOF COIL CHAIN HG 801313																	
1/4"X800FT	110	88000FT	7	31.5	9.8	5200	2600	1300	15	0.13	0.14	0.38	0.022	0.035			
1/4"X400FT	50	20000FT	7	31.5	9.8	5200	2600	1300	15	0.13	0.14	0.38	0.022	0.035			
G100 ALLOY CHAIN BLACK 801313																	
5/16"X500FT	1	500FT	8	26.4	11.5	18000	9000	4500	15	0.21	0.21	1.01	0.008	0.002			
STAINLESS STEEL CHAIN 304 801313																	
3/8"X400FT	10	4000FT	10	35	14	16200	8100	5400	15	0.034	0.32	0.7	0.034	0.001	18.2	8.05	
7/8"X80FT	2	160FT	22	77	27.5	73500	36750	24500	15	0.034	0.32	0.7	0.034	0.001	18.2	8.05	

INSPECTION DATE Dec 15 2017
NANTONG HEXING CHAIN CO.,LTD.

NANTONG HEXING CHAIN CO.,LTD.

ADD: XINDIAN INDUSTRIAL ZONE,RUDONG COUNTY, NANTONG CITY, JIANGSU PROVINCE, CHINA

TO:ELITE SALES,INC,

PO Number:801280/801298/801313

LC Number: 237

PHYSICAL AND CHEMICAL MILL TEST CERTIFICATES

DESCRIPTION OF GOODS	PCS	TOTAL LENGTH	SIZE			TENSILE TEST				CHEMICAL COMPOSITION							
			D mm	L mm	b mm	BREAKING LOAD(LBS)	PROOF LOAD(LBS)	WORKING LOAD(LBS)	ELONGATIC %	C	Si	Mn	P	S	Cr	Ni	Mo
G30 PROOF COIL CHAIN EG 801280																	
3/16"X250FT	70	17500FT	5.5	24.38	7.7	3200	1600	800	15	0.13	0.14	0.38	0.022	0.035			
5/16"X550FT	20	11000FT	8	32.8	11.2	7600	3800	1900	15	0.13	0.14	0.38	0.022	0.035			
G30 PROOF COIL CHAIN HG 801280																	
3/16"X1500FT	10	15000FT	5.5	24.8	7.7	3200	1600	800	15	0.13	0.14	0.38	0.022	0.035			
7/8"X80FT	10	800FT	22	77	27.5	51200	25600	12800	15	0.13	0.14	0.38	0.022	0.035			
G70 BINDING CHAIN YZP 801280																	
5/16"X20FT	50	1000FT	8.7	32.8	11.2	18800	9400	4700	15	0.21	0.32	1.55	0.017	0.001			
G80 ALLOY CHAIN BLACK 801280																	
1/2"X200FT	25	5000FT	13	41.5	18.5	48000	24000	12000	15	0.21	0.32	1.55	0.017	0.001			
G70 TRANSPORT CHAIN YZP 801280																	
1/4"X800FT	25	20000FT	7	31.5	9.8	12600	6300	31500	15	0.21	0.32	1.55	0.017	0.001			
STAINLESS STEEL CHAIN 316 801280																	
1/8"X1500FT	5	7500FT	4	23.9	6.4	2400	1200	600	15	0.034	0.32	0.7	0.034	0.001	18.2	8.05	
G80 ALLOY CHAIN BLACK 801298																	
1/2"X200FT	50	10000FT	13	41.5	18.5	48000	24000	12000	15	0.21	0.32	1.55	0.017	0.001			
G30 PROOF COIL LONG CHAIN HG 801313																	
1/2"X200FT	14	2800FT	13	88.9	20.27	18000	9000	4500	15	0.13	0.14	0.38	0.022	0.035			
1"X100FT	5	500FT	25	104.78	34.93	71600	35800	17900	15	0.13	0.14	0.38	0.022	0.035			
5/8"X100FT	7	700FT	16	95.25	25.4	27600	13800	6900	15	0.13	0.14	0.38	0.022	0.035			
G43 BOOMER CHAIN SC 801313																	
1/2"X20FT	100	2000FT	13	45.5	18.2	27600	13900	9200	15	0.21	0.32	1.55	0.017	0.001			
G30 PROOF COIL CHAIN HG 801313																	
1/4"X800FT	110	88000FT	7	31.5	9.8	5200	2600	1300	15	0.13	0.14	0.38	0.022	0.035			
1/4"X400FT	50	20000FT	7	31.5	9.8	5200	2600	1300	15	0.13	0.14	0.38	0.022	0.035			
G100 ALLOY CHAIN BLACK 801313																	
5/16"X500FT	1	500FT	8	26.4	11.5	18000	9000	4500	15	0.21	0.21	1.01	0.008	0.002			
STAINLESS STEEL CHAIN 304 801313																	
3/8"X400FT	10	4000FT	10	35	14	16200	8100	5400	15	0.034	0.32	0.7	0.034	0.001	18.2	8.05	
7/8"X80FT	2	160FT	22	77	27.5	73500	36750	24500	15	0.034	0.32	0.7	0.034	0.001	18.2	8.05	

INSPECTION DATE Dec 15 2017
NANTONG HEXING CHAIN CO.,LTD.

NANTONG HEXING CHAIN CO.,LTD.

ADD: XINDIAN INDUSTRIAL ZONE,RUDONG COUNTY, NANTONG CITY, JIANGSU PROVINCE, CHINA

TO:ELITE SALES,INC,

PO Number:801280/801298/801313

LC Number: 237

PHYSICAL AND CHEMICAL MILL TEST CERTIFICATES

PHYSICAL AND CHEMICAL MILL TEST CERTIFICATES										CHEMICAL COMPOSITION							
DESCRIPTION OF GOODS	PCS	TOTAL LENGTH	SIZE			TENSILE TEST				CHEMICAL COMPOSITION							
			D mm	L mm	b mm	BREAKING LOAD(LBS)	PROOF LOAD(LBS)	WORKING LOAD(LBS)	ELONGATIO %	C	Si	Mn	P	S	Cr	Ni	Mo
G30 PROOF COIL CHAIN EG 801280																	
3/16"X250FT	70	17500FT	5.5	24.38	7.7	3200	1600	800	15	0.13	0.14	0.38	0.022	0.035			
5/16"X550FT	20	11000FT	8	32.8	11.2	7600	3800	1900	15	0.13	0.14	0.38	0.022	0.035			
G30 PROOF COIL CHAIN HG 801280																	
3/16"X1500FT	10	15000FT	5.5	24.8	7.7	3200	1600	800	15	0.13	0.14	0.38	0.022	0.035			
7/8"X80FT	10	800FT	22	77	27.5	51200	25600	12800	15	0.13	0.14	0.38	0.022	0.035			
G70 BINDING CHAIN YZP 801280																	
5/16"X20FT	50	1000FT	8.7	32.8	11.2	18800	9400	4700	15	0.21	0.32	1.55	0.017	0.001			
G80 ALLOY CHAIN BLACK 801280																	
1/2"X200FT	25	5000FT	13	41.5	18.5	48000	24000	12000	15	0.21	0.32	1.55	0.017	0.001			
G70 TRANSPORT CHAIN YZP 801280																	
1/4"X800FT	25	20000FT	7	31.5	9.8	12600	6300	31500	15	0.21	0.32	1.55	0.017	0.001			
STAINLESS STEEL CHAIN 316 801280																	
1/8"X1500FT	5	7500FT	4	23.9	6.4	2400	1200	600	15	0.034	0.32	0.7	0.034	0.001	18.2	8.05	
G80 ALLOY CHAIN BLACK 801298																	
1/2"X200FT	50	10000FT	13	41.5	18.5	48000	24000	12000	15	0.21	0.32	1.55	0.017	0.001			
G30 PROOF COIL LONG CHAIN HG 801313																	
1/2"X200FT	14	2800FT	13	88.9	20.27	18000	9000	4500	15	0.13	0.14	0.38	0.022	0.035			
1"X100FT	5	500FT	25	104.78	34.93	71600	35800	17900	15	0.13	0.14	0.38	0.022	0.035			
5/8"X100FT	7	700FT	16	95.25	25.4	27600	13800	6900	15	0.13	0.14	0.38	0.022	0.035			
G43 BOOMER CHAIN SC 801313																	
1/2"X20FT	100	2000FT	13	45.5	18.2	27600	13900	9200	15	0.21	0.32	1.55	0.017	0.001			
G30 PROOF COIL CHAIN HG 801313																	
1/4"X800FT	110	88000FT	7	31.5	9.8	5200	2600	1300	15	0.13	0.14	0.38	0.022	0.035			
1/4"X400FT	50	20000FT	7	31.5	9.8	5200	2600	1300	15	0.13	0.14	0.38	0.022	0.035			
G100 ALLOY CHAIN BLACK 801313																	
5/16"X500FT	1	500FT	8	26.4	11.5	18000	9000	4500	15	0.21	0.21	1.01	0.008	0.002			
STAINLESS STEEL CHAIN 304 801313																	
3/8"X400FT	10	4000FT	10	35	14	16200	8100	5400	15	0.034	0.32	0.7	0.034	0.001	18.2	8.05	
7/8"X80FT	2	160FT	22	77	27.5	73500	36750	24500	15	0.034	0.32	0.7	0.034	0.001	18.2	8.05	

INSPECTION DATE Dec 15 2017
NANTONG HEXING CHAIN CO.,LTD.

SECTION - 3.C

Door Gasket

Custom Rubber Extrusion Division

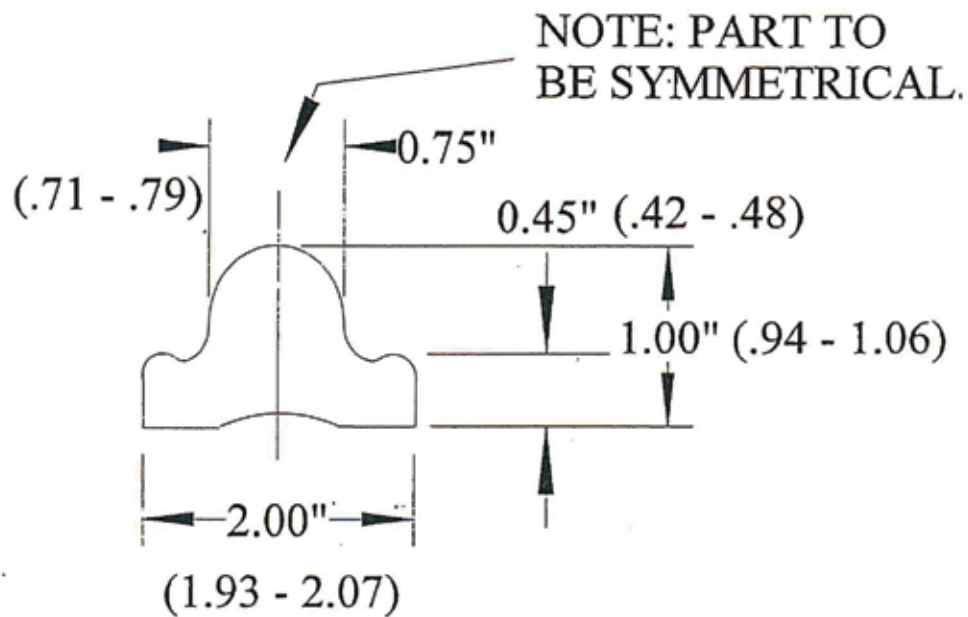
EXTRUSION CAPABILITIES: 3 Movable extruders ranging from 3-1/2" to 6"

Compounds:	<u>Dense Rubber</u> - SBR, EPDM, Nitrile, Neoprene, Natural Rubber, Polyisoprene, & Silicone. For high volume (5M+ lbs.) applications only: Butyl, Hypalon, & Epichlorohydrin. <u>Closed Cell SPONGE</u> -Neoprene, EPDM, & Silicone.
Durometers:	30 - 90 Shore A depending on compound
Colors:	Primarily Black: Some Gray, Yellow, & other colors.
Sizes:	Solid profiles: From 0.5" square to 8" square Other profiles: Up to 4" high x 18" wide Tubing: OD's from .125" up to 7.0". Minimum wall of .040"
Curing Processes:	<u>Microwave</u> -- Continuous in-line curing with <u>Laser Control System</u> . "Clean" Profiles with slight markings from conveyor belt on one surface. Unlimited lengths available - depending on profile and packaging. <u>Steam - 6" Cold feed Extruder</u> for <u>Large Parts</u> up to 55 ft. long & <u>14 lb./ft.</u> Profiles will have slight evidence of white talc on them due to handling. <u>Salt</u> - Continuous in-line curing for small to wide parts with no marking on extrusions & round tubing. Also capable of color extrusions.
Quantities:	In general, a larger volume manufacturer. Minimum quantities will be identified based on profile size, compound selection, and processing equipment required.

ADDITIONAL NOTES:

Other capabilities:	Splicing and Corner Molding, Form Curing, Cutting-to-Length, Angle Cuts, Hole Punching + other secondary operations. Custom Compound Mixing & Development.
"Not a Good Fit":	Thermoplastics (Flexible PVC, Santoprene), Dual Durometers, fabric/metal reinforced hoses.

"T-GASKETS"



RHPN: 73840003

50 Durometer Nitrile

Material Specification

Compound: 840

Requirements:

Typical Properties

Durometer [Shore A]:
Tensile (psi):
Elongation (%):

45 – 55
1,500
500

Base Polymer:

Nitrile

Quality Assurance

SECTION - 3.D

Fasteners



VIRAJ PROFILES LIMITED (FASTENER DIVISION)

TARAPUR, MAHARASTRA, INDIA

MATERIAL TEST REPORT (CERTIFICATION TO EN-ISO 10204-3.1)

CUSTOMER : BRIGHTON-BEST INTERNATIONAL (TAIWAN) INC NO. 122,YILIN RD, RENDE TOWNSHIP TAIWAN COUNTY,717 TAIWAN ,R.O.C	MANUFACTURER'S	INVOICE NUMBER / MTR NO.		52-980433/24		
	SYMBOL	DATE		05.11.2014		
	VIRAJ OR 	ITEM DESCRIPTION		Stainless Steel 316 (Inch) Hex Nuts		
		SIZE		3/8" - 16		
		TOTAL QUANTITY		96000 Pieces	PART NO	763036
		MATERIAL SPECIFICATION		ASTM F594	LOT NO	041114002492
		CONDITION		CW1		
DIMENSIONAL SPECIFICATION		ASME B 18.2.2				

CHEMICAL ANALYSIS

Sr.No	VALUES		Heat No	%C	%Mn	%Si	%S	%P	%Cr	%Ni	%Mo	%N2	Conforms to ASTM F594 Alloy Group-2 SS3316
	Required Values	Min											
		Max											
1	Observed Values		34262	0.017	1.35	0.37	0.005	0.036	16.85	10.14	2.01	0.035	

DIMENSIONAL REPORT / MECHANICAL PROPERTIES

Sr.No	Characteristic	Unit	Instrument	Required		No. of Sample Inspected	Observed Values		
				Min	Max		1	2	Sample Determinant
1	Width Across Flats	mm	Micrometer	14.00	14.27	As per lot	14.06	14.26	Qualified
2	Total Thickness	mm	Micrometer	8.13	8.55	As per lot	8.23	8.48	Qualified
3	Minor Diameter	mm	Micrometer	7.80	8.15	As per lot	7.85	7.96	Qualified
4	Flange Or Collar Diameter	mm	Micrometer	##NA	##NA	As per lot	##NA	##NA	Qualified
5	Gauge Fit / Thread Specification		Gauge	As per ASME B1.1		100 Nos	2B Go Answering freely. 2B No Go Not answering		
6	Proof Load	N	Tensile Testing M/c	34472		2 Nos	34558	34589	Qualified
7	Hardness	HRB	Hardness Tester	95HRB	106HRB	3 Nos	102	101	101
8	Marking	Visual		Logo Or VIRAJ	F594G 316	100 Nos	OK	OK	OK
9	Surface Discontinuities	Visual		As per ASTM F812-2002		100 Nos	No surface discontinuities observed		
10	Appearance	Visual				100 Nos	Satisfactory		
11	Surface Treatment		Passivation				Chemical Passivated and Conforms to ASTM A967-05(practice-A)		

Remarks :

DIMENSIONAL INSPECTION : CONFIRM TO THE SPECIFICATION

SURFACE INSPECTION : CONFIRM TO THE SPECIFICATION

USED WIRE ROD : FROM CONCAST BILLET

NA = NOT APPLICABLE

WE CERTIFY THAT THE MATERIAL DESCRIBED ABOVE HAS BEEN TESTED AND COMPLIES WITH THE ABOVE MENTIONED SPECIFICATION AND IS OF INDIAN ORIGIN AND FREE FROM RADIOACTIVE CONTAMINATION

Authorised Signatory - QA

TONG HWEI ENTERPRISE CO., LTD.
NO.399, GANGYAN ROAD, GANGSHAN 820, KAOHSIUNG TAIWAN, R.O.C. TEL: 886-7-6222851 FAX: 886-7-6211234

CERT OF INSPECTION

Cert no.: THB3060063-329 Page 1 of 1
 STAR STAINLESS SCREW CO.
 30 WEST END ROAD
 TOTOWA NJ 07511
 U. S. A.

Item no.: 32-FS-25C-125
 Po# no.: QQQ120320JUN
 Inv. no.: THB3060063
 Issue date: Jun 26, 2013
 Insp. date: Apr 07, 2012

Lot no.: B2407037
 Ship qty: 20,800 PCS
 Lot qty: 48,472 PCS
 Sampler: H.G. TSAI
 Sample plan spec.: ANSI/ASME B18.18.2M-1987
 Mfr.: TONG HWEI ENTERPRISE CO., LTD.
 Address: SAME AS LAB.
 Mfr. date: Mar 28, 2012

SS 316CU; SKT FLAT CSK HEAD C/S 82+/-2, UNC; FULL THREAD; PASSIVATED; MARK "316" ON THE TOP HEAD
 1/4-20 X 1-1/4 UNC 3A

Characteristic	Requirements	Results	S/S	Pass	Rej	Specification	Test Method
Dimensional inspections (Measurement by millimeters)							
Head Diameter	12.20 - 13.48	12.29 - 12.41	8	8	0	ANSI/ASME B18.3-02	
Nominal Length	30.73 - 31.75	31.16 - 31.22	32	32	0	ANSI/ASME B18.3-02	
Head Height	MIN 4.09	4.32 - 4.33	8	8	0	ANSI/ASME B18.3-02	
Socket Width Across Flats	3.97 - 4.03	PASSED	8	8	0	ANSI/ASME B18.3-02	
Major Diameter	6.150 - 6.350	6.151 - 6.198	32	32	0	ANSI/ASME B18.3-02	
Key Engagement	MIN 2.82	3.10 - 3.12	8	8	0	ANSI/ASME B18.3-02	
Pitch Diameter	5.460 - 5.520	5.470 - 5.480	32	32	0	ANSI/ASME B18.3-02	
Protrusion	1.10 - 1.62	1.47 - 1.48	8	8	0	ANSI/ASME B18.3-02	
Visual Appearance		PASSED	100	100	0	ASTM F788-08	
Mechanical properties							
Core Hardness (HRBW)	80 - 100	94 - 96	8	8	0	CUSTOMER REQUIREMENT	ASTM F606-11a
Tensile Strength (KSI)	80MIN	109.4 - 110.1	4	4	0	CUSTOMER REQUIREMENT	ASTM F606-11a
Yield Strength(0.2% offset)							N/A

Chemical composition %(Acc. to Material Inspection Certificate) Heat No.: 3S228										Material Source: TAIWAN				
C	Si	Mn	P	S	Ni	Cr	Mo	Cu	N					
X100	X100	X100	X1000	X1000	X100	X100	X100	X100	ppm					
1.4	30	83	26	0.2	1050	1651	205	305						

Remark: * TWO INDENTATION ONLY INSTEAD OF FOUR ON CORE HARDNESS.

All hardness requirements are consistent with THE standards if hardness requirements not provided to specification above

1. This inspection certificate is issued according to EN 10204 3.1/DIN 50049 3.1.
2. This report must not be reproduced except in full and to the item(s) tested only.
3. Whenever testing discrepancies are detected or departures from documented policies and procedure occur, they will be shown in remark.

Y. L. Su
 Y. L. SU (QA ENGINEER)

TONG HWEI ENTERPRISE CO., LTD.
 NO.399, GANGYAN ROAD, GANGSHAN 820, KAOHSIUNG TAIWAN, R.O.C. TEL: 886-7-6222851 FAX: 886-7-6211234

CERT OF INSPECTION

Cert no.: THB5050018-834 Page 1 of 1
 STAR STAINLESS SCREW CO.
 30 WEST END ROAD
 TOTOWA NJ 07511
 U. S. A.

Item no.: 32-FS-25C-150
 Po# no.: QQQ132175MAY
 Inv. no.: B5050018
 Issue date: May 25, 2015
 Insp. date: Dec 09, 2013

Lot no.: B3C09083
 Ship qty: 16,800 PCS
 Lot qty: 27,029 PCS
 Sampler: H.C. JIN

Mfr.: TONG HWEI ENTERPRISE CO., LTD.
 Address: SAME AS LAB.
 Mfr. date: Nov 25, 2013

SS 316CU; SKT FLAT CSK HEAD C/S 82+/-2, UNRC; FULL THREAD; PASSIVATED; MARK "316" ON THE TOP HEAD
 1/4-20 X 1-1/2

Characteristic	Requirements	Results	S/S	Pass	Rej	Specification	Test Method / [Device*1] / (Sample Plan*2)							
Dimensional inspections														
Head Diameter (in)	0.480 - 0.531	0.4854 - 0.4858	15	15	0	ASME B18.3-2003 ASME B18.3-2003 ASME B18.3-2003 ASME B18.3-2003 ASME B18.3-2003 ASME B18.3-2003 ASME B1.1-2003 ASME B1.1-2003 ASTM F788-13	[Cal] (18)							
Socket Width Across Flats (in)	0.156 - 0.159	0.1567 - 0.1571	4	4	0		[Cal] (18)							
Gauge of hexagon sockets	5/32	PASSED	15	15	0		[Hex] (18)							
Key Engagement (in)	MIN 0.111	0.1189 - 0.1201	15	15	0		[Hep] (18)							
Protrusion (in)	0.043 - 0.064	0.0630 - 0.0630	15	15	0		[Hhg] (18)							
Nominal Length (in)	1.4600 - 1.5000	1.4969 - 1.4980	15	15	0		[Cal] (18)							
Major Diameter (in)	0.2419 - 0.2500	0.24646 - 0.24665	15	15	0		[Mcr] (18)							
Pitch Diameter (in)	0.2147 - 0.2175	0.21535 - 0.21575	15	15	0		[Tri] (18)							
Visual Appearance		PASSED	29	29	0		(18)							
Mechanical properties														
Core Hardness*3 (HRBW)	MIN 50 HRA	96 - 97	15	15	0	ASTM F879-12 ASTM F879-12	ASTM F606-14 [Rht] (14) ASTM F606-14 [Mtm] (14)							
Tensile Strength (KSI)	MIN 80.0	122.4 - 123.1	4	4	0									
Chemical composition % (Acc. to Material Inspection Certificate) Heat No.: 4E530														
C	Si	Mn	P	S	Ni	Cr	Mo	Cu	N	Material Source: TAIWAN				
X100	X100	X100	X1000	X1000	X100	X100	X100	X100	X10000					
4.2	32	120	28	2.7	1001	1610	204	36	272					
Remark:														
*1 Device: [Cal] Vernier caliper [Hep] Hexagon Penetration Gauge [Hhg] Hexagon Gage														

Remark:

- *1 Device: [Cal] Vernier caliper [Hep] Hexagon Penetration Gauge [Hex] Hexagon Plug gauge [Hhg] Protrusion-Head Height Gage [Mcr] Micrometer [Mtm] Material Testing Machine [Rht] Rockwell Hardness Tester [Tri] Tri-Roll Thread Comparator
 *2 Sample Plan: (14) ASTM F1470-2012 MECHANICAL TEST (18) ASME B18.18-2011 CATEGORY 2
 *3 Two indentation only

- This inspection certificate is issued according to EN 10204 3.1/DIN 50049 3.1.
- This report must not be reproduced except in full and to the item(s) tested only.
- Whenever testing discrepancies are detected or departures from documented policies and procedure occur, they will be shown in remark.
- If the products are needed to passivate, we confirm that all are passivated with slight liquid wax.
- Approximate Rockwell Hardness Conversion Numbers From HRC or HRBW to HRA for Austenitic Stainless Steel, please refer to ASTM E140 Table 6.

834 of 918

Y. L. Su
 Y. L. SU (QA ENGINEER)

AC051173

yuan 09:33:40

TONG HWEI ENTERPRISE CO., LTD.

NO.399, GANGYAN ROAD, GANGSHAN 820, KAOHSIUNG TAIWAN, R.O.C. TEL: 886-7-6222851 FAX: 886-7-6211234

CERT OF INSPECTION

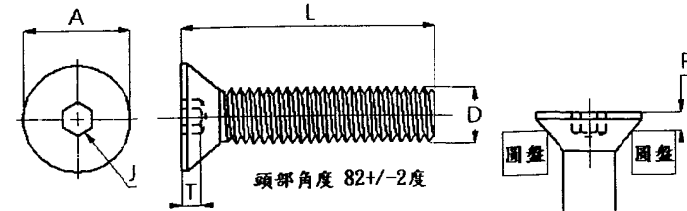
Cert no.: THB6040107-764 Page 1 of 1

STAR STAINLESS SCREW CO.
30 WEST END ROAD
TOTOWA NJ 07511
U. S. A.

SS 316CU; SKT FLAT CSK HEAD C/S 82+/-2, UNRC; FULL
THREAD; PASSIVATED; MARK "316" ON THE TOP HEAD

3/8-16 X 1-1/4

Lot no.: B6307184
Item no.: 32-FS-37C-125
Po# no.: QQQ138409MAY
Inv. no.: B6040107
Issue date: Apr 29, 2016
Insp. date: Mar 07, 2016
Ship qty: 20,000 PCS
Lot qty: 26,082 PCS
Sampler: H.G. TSAI
Mfr/Addr: Same as Lab
Mfr. date: Mar 04, 2016



Characteristic	Requirements	Results	S/S	Pass	Rej	Specification ^{*1}	Test Method [Device ^{*2}] (Sample Plan ^{*3})
Dimensional inspections							
A Head Diameter (in)	0.720 - 0.781	0.7205 - 0.7280	15	15	0	ASME B18.3-2012	[Cal] (18)
J Socket Width Across Flats (in)	0.219 - 0.222	0.2209 - 0.2213	4	4	0	ASME B18.3-2012	[Cal] (18)
GJ Gauge of hexagon sockets	7/32	PASSED	15	15	0	ASME B18.3-2012	[Hex] (18)
T Key Engagement (in)	MIN 0.159	0.1732 - 0.1748	15	15	0	ASME B18.3-2012	[Hep] (18)
P Protrusion (in)	0.050 - 0.088	0.0756 - 0.0760	15	15	0	ASME B18.3-2012	[Hhg] (18)
L Nominal Length (in)	1.2100 - 1.2500	1.2331 - 1.2343	15	15	0	ASME B18.3-2012	[Cal] (18)
D Major Diameter (in)	0.3656 - 0.3750	0.37177 - 0.37209	15	15	0	ASME B1.1-2003	[Mcr] (18)
PD Functional Dia.	UNRC 3A	PASSED	15	15	0	ASME B1.1-2003	[Trg] (18)
PD Pitch Diameter (in)	0.331 - 0.334	0.33110 - 0.33157	15	15	0	ASME B1.1-2003	[Tri] (18)
VA Visual Appearance		PASSED	29	29	0	ASTM F788-13	(18)
Mechanical properties							
HR Core Hardness ^{*4} (HRBW)	MIN 50 HRA	98 - 100	15	15	0	ASTM F879-12	ASTM F606-14 [Rht] (14)
TS Tensile Strength (KSI)	MIN 80.0	109.9 - 110.8	4	4	0	ASTM F879-12	ASTM F606-14 [Mtm] (14)
EL Extension (in)	MIN 0.150	0.1646 - 0.1732	4	4	0	ASTM F879-12	ASTM F606-14 [Cal] (14)

Chemical composition % (Acc. to Material Inspection Certificate)					Heat No.: 4T430	Material Source: TAIWAN							
C	Si	Mn	P	S	Ni	Cr	Mo	Cu	N				
X100	X100	X100	X1000	X1000	X100	X100	X100	X100	X10000				
1.9	29	80	27	0.8	1053	1662	203	304	241				

Remark:

*1 Additional explain: N/A

*2 Device: [Cal]Vernier caliper [Hep]Hexagon Penetration Gauge [Hex]Hexagon Plug gauge [Hhg]Protrusion-Head Height Gage [Mcr]Micrometer [Mtm]Material Testing Machine [Rht]Rockwell Hardness

Tester [Trg]Thread Ring Gauge [Tri]Tri-Roll Thread Comparator

*3 Sample Plan: (14)ASTM F1470-2012 MECHANICAL TEST (18)ASME B18.18-2011 CATEGORY 2

*4 Three indentation only

- This inspection certificate is issued according to EN 10204 3.1/DIN 50049 3.1. It must not be reproduced except in full and to the item(s) tested only.
- Product meets the order requirements.
- Whenever testing discrepancies are detected or departures from documented policies and procedure occur, they will be shown in remark.
- If the products are needed to passivate, we confirm that all are passivated with slight liquid wax. Passivation test per ASTM A380-2006: Passed.
- Approximate Rockwell Hardness Conversion Numbers From HRC or HRBW to HRA for Austenitic Stainless Steel, please refer to ASTM E140 Table 6.

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Y. L. SU (QA ENGINEER)

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SECTION - 3.E

Filter Media



PPFM-62-350



Mesh (per inch)	68 X 20
Description (weave)	6/2 Satin
Yarn Type:	Polyester / Polyester
Warp / Weft	0.0197" / 0.0315 (500 microns / 800 microns)
Thickness	0.085"
Weight	42.5 oz/yd ²
Air Permeability	325 – 355 CFM
Water Permeability	345 mm ³ / mm ² / sec
Tensile Strength	1560 lbs/inch
Micron Opening	0 X 325
Micron Retention	260
Open Area	19%
Available Seams	316 SS Clipper

SECTION - 3.F

Rubber Gaskets



STYLE 759CCE

Style 759CCE is a closed cell Nitrile PVC medium density sponge gasket material. Excellent material for applications including gasket, seals, insulation, vibration and noise dampening.

Type	IV2
Polymer	NITRILE
Color	Black
ASTM D-1056 Classification	2B2
25% Compression Resistance (psi)	5.0 – 9.0
50% Compress Set (%)	40
Density (lb. /ft2)	5.5 – 7.5
Water Absorption (lb. /ft2)	0.1 max.
Tensile (psi)	75 min.
Elongation (%)	100 min.
Flammability – MVSS302	Pass
UL 94	Pass



250 Chester Street
Painesville, Ohio 44077
 Phone: **866-GO-AVERY (866-462-8379)**
 Fax: **888-358-4469**

email: psa.tape@averydennison.com

URL: stus.averydennison.com

Surface Preparation

It is essential, as with all pressure-sensitive tapes, that the surface to which the tape is applied be clean, dry, and free of grease or oil.

Temperatures

Min Application Temp	50° F	10° C
Max Continuous Operating Temp	150° F	66° C
Max Intermittent Operating Temp	175° F	79° C

Storage and Shelf Life

One year when stored at 70°F (21°C) 50% RH out of direct

PRODUCT DESCRIPTION

Applications

Designed to provide an economical product for general purpose bonding.

Features

Polyester carrier
 Excellent tack and adhesion
 Densified kraft release liner

Benefits

Excellent processing and die-cutting stability
 Suited for a broad range of general purpose applications

PRODUCT DATA

Product Construction

Test Methods: PSTC-133

		US Mils	MM's
Unwind Adhesive	Rubber Based	2.0	0.05
Carrier	Polyester	0.5	0.01
Liner Adhesive	Rubber Based	1.5	0.04
Liner	80 # Densified Kraft	4.6	0.12
	Total Construction	8.6	0.22

Adhesive Properties

Peel Adhesion

Test Method(s): PSTC-103, ASTM D-3330, STD-10

2 mil PET 180° 12" / min Initial				
Substrate	Test Side	Lbf / in	US Oz / in	N / Meter
SS	Liner	9.5	152	1665

ABS	Liner	8.3	132	1450
------------	-------	------------	------------	-------------

PP	Liner	6.6	105	1152
-----------	-------	------------	------------	-------------

Talc PP	Liner	2.5	40	436
----------------	-------	------------	-----------	------------

LDPE	Liner	4.4	70	770
-------------	-------	------------	-----------	------------

HDPE	Liner	4.5	73	795
-------------	-------	------------	-----------	------------

Painted Metal	Liner	8.8	140	1537
----------------------	-------	------------	------------	-------------

Static Shear

Test Method(s): PSTC-107, ASTM D 3654, STD-9

2 mil PET @ 72°F / 22°C 1" sq (6.5 cm2) 5.5 lbs (2.5 kg)	
Substrate	Test Side
SS	Liner
	Min to Fail
	> 1,200

Limited Warranty

All statements, technical information and recommendations concerning products sold or samples provided by AVERY DENNISON are based upon tests believed to be reliable but do not constitute a guarantee or warranty. All products are sold and samples of products provided with the understanding that PURCHASER has independently determined the suitability of such products for its purposes. AVERY DENNISON warrants the products to be free from defects in material and workmanship. Should any failure to conform to this warranty appear within one year* after the initial date of shipment, AVERY DENNISON shall, upon notification thereof and substantiation that the products have been stored and applied in accordance with AVERY DENNISON's standards, correct such defects by suitable repair or replacement without charge at AVERY DENNISON's plant or at the location of the products (at AVERY DENNISON's election) provided, however, if AVERY DENNISON determines that the repair or replacement is not commercially practical, AVERY DENNISON shall issue credit in favor of PURCHASER in an amount not to exceed the purchase price of the products.

This warranty is exclusive and is in lieu of any implied warranty of merchantability, fitness for a particular purpose or other warranty of quality, whether express or implied, except the warranty of title and against patent infringement. No waiver, alteration, additions or modifications of the foregoing conditions shall be valid unless made in writing and manually signed by an officer of Avery Dennison. *Or in the time period stated on the specific product specification sheet, if any, and if not then on the specific product information literature in effect at time of shipment.

Limitation of Liability

In no event shall AVERY DENNISON be liable for any incidental or consequential damages, including but not limited to, loss of profit, loss of use of production or loss of capital. The remedies of PURCHASER set forth herein are exclusive and the total liability of AVERY DENNISON with respect to any contract, or anything done in connection therewith such as the performance or breach hereof, or from the manufacture, sale, delivery, resale, installation or use of products whether arising out of contract, negligence, strict tort, or under any warranty, or otherwise, shall not exceed the purchase price of the products upon which the liability is based.

SECTION - 3.G

Tarp



22 oz VINYL COATED POLYESTER MATERIAL

ITEM N2 PC-22

BASE FABRIC	
Weight	6 oz PSY
Fiber	Polyester (1300D x 1300D)
Fabric Style	Woven (18 x 18)

COATED FABRIC	
Total Weight	22 +/- 0.5 oz PSY
Type of Coating	Poly Vinyl Chloride
Coating Distribution	60% Front / 40% Back
Width	126"
Sealing Properties	[X] Dielectric [X] Thermal

MECHANICAL PROPERTIES		RESULTS	TEST METHODS	
			Fed. Std. 19A	ASTM
Grab Tensile:	Warp	430 Lbs	5100	D5034
	Fill	410 Lbs		
Strip Tensile:	Warp (1")	320 Lbs	5102	D5035
	Fill (1")	310 Lbs		
Tongue Tear:	Warp	110 Lbs	5134	0751
	Fill	100 Lbs		
Adhesion:	Adhesive (1")	10 Lbs		
Abrasion (Taber):	Wheel: H-18	1000 Cycle, 70 rpm	5106	03884
	Load: 1000 gr			
Hydro Resistance:		600 PSI	5512	
Low Temperature:		-40 degrees F		
Flame Resistance:	Time of Afterflame	N/A	N/A	N/A
	Length of Char	N/A	N/A	N/A

The information provided represents typical physical properties but does not constitute a specification. See remarks for no representation of warranty, expressed or implied. The products shown are subject to change without notice of the Buyer and the product will be property of the Buyer for the buyer's purpose. The buyer, solely, is responsible for the proper use of this product for the intended use of the Buyer.



TARP STRAPS • (NR) RUBBER - Sri Lanka

PHYSICAL PROPERTIES;

TEST:	STANDARD:	MILL REQUIREMENT:	PERFORMANCE:
Tensile Strength	(ASTM D-412)	160 kg/cm ² (2275 psi)	208 kg/cm ² (2550 psi)
Load @Break	(No ASIM Std)	None	120 Kg (264.5 lbs)
Elongation @50lbs	(No ASTM Std)	100 + or - 5%	100 %
Elongation @Break	(ASTM D-412)	350%	500 %
Modulus @300%	(No ASTM Std)	62 kg/cm ² (880 psi)	77.2 kg/cm ² (1095 psi)
Durometer	(ASTM D-2240)	55+ OR- 3% IRHD	58 IRHD
Stretch Recovery	(No ASTM Std)	80-100%	98.8%
Specific Gravity	(No ASTM Std)	None	1.17
Low Temperature	(ASTM D-2137-94)	PASS	Pass (-51 degrees C)
Ozone Resistance	(ATSM D-1149-91)	Pass	Pass

SPECIFICATION FOR THE "S" HOOKS:

4.4 mm (0.18 inch) Diameter. Silicon-Kilned Hard Temper Galvanized (Min 45 g/m²) Wire
2.5 Inch Height. Min 65 kg/mm² (92,000 psi) Tensile Strength

DEFINITIONS / TERMINOLOGY:

Tensile Strength	The maximum pulling force required to cause failure of a specified specimen, the force acting parallel to the major axis of the test specimen.
Load @ Break	The amount of pulling force applied to cause failure of an individual tarp strap.
Elongation	The length or stretch at a specific: amount of pull divided by the: original length multiplied by 100%.
Elongation @Break	The length of the specimen at maximum stretch before failure divided by the original length multiplied by 100% {Do Not stretch beyond the Maximum Stretch printed on each strap while in application or service}
Modulus of Elasticity	The ratio of change in stress to change in strain for stresses below the elastic limit. A lower value is an indication of a more elastic specimen or compound. In this case the value indicates the ease or difficulty in stretching the tarp strap.
Durometer	The hardness or resistance to indentation of the rubber compound (toughness).
Stretch Recovery	Determines of percent of recovery <i>after</i> being stretched 150% of original length.
Specific Gravity	The ratio of the mass of a material to the mass of an equal volume of water. Measure of the density of the strap. The higher the specific gravity, the more the strap contains fillers in the composition.
Low Temperature	Determination of brittlepoint. Remain Non-Brittle after 3 minutes at -40 degrees C.
Ozone Resistance	Evaluates weatherability or ozone deterioration of a vulcanized specimen when exposed to an atmosphere containing Ozone. The specimen is stretched to a specified tensile strain and exposed to 50 PPHM Ozone Concentration for 24 hours at 40 degrees c (104 degrees F). Exposure time and rating under 7x magnification.

SECTION - 4

Paint Specifications

SELECTION & SPECIFICATION DATA

Generic Type	Phenolic epoxy novolac
Description	This product is a solvent-free, high performance epoxy coating designed as an internal tank, valve and pipe lining for chemical or other commodity storage. It is a unique blend of resins and curing agents that allow batch mixing for ease of application. Plural component spray equipment is not required. The product is blush resistant and is typically applied at film thicknesses of 20 mils (500 microns) or thicker as needed (tank floors). It can handle exposures typically seen in the oil and gas industries; crude oils and fuels. It is resistant to NGL condensates, produced water, brines, industrial process water, waste water, and sewage. Ideal for municipal wastewater and water treatment facilities. Phenoline Tank Shield may be used as a patching and seam sealer by adding Carboline Thixatropo D to the pre-mixed kit at a ratio between 2:1 and 1:1 by volume resin:powder.
Features	• Batch mix formulation, single leg airless spray • High impact resistance • Superior adhesion to steel • Excellent resistance to water and salt water • Resistance to a broad range of fuels • Passes EI 1541 requirements for internal protective coating systems used in aviation fuel handling systems, including the jet fuel gum test • Adheres to API 653 inspection and API 652 guidelines for inspection intervals. • Resistant to hot water up to 180°F (82°C) • Excellent abrasion resistance and flexibility • Can be applied down to 35 °F (2 °C) • Can be applied as a single or multi-coat system • Non-blushing with a long recoat window • Low odor
Color	Standard: Grey (N700), White (N800), Blue (N100)
Finish	Gloss
Primer	Coating is normally applied direct to metal. May be applied over other primers as recommended by Carboline.
Dry Film Thickness	12 - 30 mils (305 - 762 microns) per coat Depends on service and existing condition of the substrate, product is typically applied in a one coat application at the appropriate film thickness depending on the application. Higher film thicknesses (60+ mils/1500+ microns) are used for more aggressive or abrasive conditions or for severely pitted steel (tank bottoms). Maximum vertical film build is 30 mils (750 microns).
Solids Content	By Volume 99% +/- 1%
Theoretical Coverage Rate	1588 ft²/gal at 1.0 mils (39.0 m²/l at 25 microns) 132 ft²/gal at 12.0 mils (3.2 m²/l at 300 microns) 53 ft²/gal at 30.0 mils (1.3 m²/l at 750 microns) Allow for loss in mixing and application.
VOC Values	As Supplied : 9 g/l
Wet Temp. Resistance	Immersion temperature resistance depends upon exposure, consult Carboline Technical Service for specific information

Phenoline[®] Tank Shield

PRODUCT DATA SHEET



SUBSTRATES & SURFACE PREPARATION

General	Surfaces must be clean and dry. Employ adequate methods to remove dirt, dust, oil and all other contaminants that could interfere with adhesion of the coating
Steel	Cleanliness: Abrasive blast to SSPC-SP10 (minimum) Profile: Minimum 3 mil (75 micron) dense, sharp anchor profile free of peening, as measured by ASTM D 4417. Defects exposed by blasting must be repaired.
Stainless Steel	Prepare by abrasive blasting to SSPC-SP 17 Thorough Abrasive Blast to a minimum of 3 mils (75 microns) dense angular anchor profile.
Concrete	Concrete: Clean and dry. Remove all loose, unsound concrete. Do not apply coating unless concrete has cured at least 28 days @ 70°F (21°C) and 50% RH or equivalent. Prepare surfaces in accordance with ASTM D4258 Surface Cleaning of Concrete and ASTM D4259 Abrading Concrete. Voids in concrete may require filling/surfacing.

MIXING & THINNING

Mixing	This product may be batch mixed and applied using standard airless spray equipment. IMPORTANT: Power mix each component separately, then combine and power mix until homogenous. Component Details for Colors: Grey (N700): The Part A is black (N909) and the Part B is white (N800) Blue (N100): The Part A is blue (N910) and the Part B is white (N800) White (N800): The Part A is clear (N000) and the Part B is white (N800)
Thinning	Thinning not normally required.
Ratio	1:1 by volume (Part A to Part B)
Pot Life	30 minutes at 75°F (24°C). Consult Carboline Technical Service for techniques to maximize pot life.

APPLICATION EQUIPMENT GUIDELINES

Listed below are general equipment guidelines for the application of this product. Job site conditions may require modifications to these guidelines to achieve the desired results.

General	This is a high solids coating and may require adjustments in spray techniques. Wet film thickness is easily and quickly achieved. The following spray equipment has been found suitable and is available from equipment manufacturers.
Airless Spray	Airless spray equipment capable of minimum 6000 psi (60:1 ratio or higher is preferred) with a minimum 180 cc lower. Fluid hose shall be minimum 3/8" I.D. Airless spray gun shall be rated minimum 7000 psi utilizing reverse-a-clean tips sizes 0.021-0.027" with fan size range between #5 to #9. A wider tip fan size facilitates break up and reduces fingering. Fixed-ratio (1:1 by volume) plural component equipment may also be used if the material cannot be sprayed within the pot life of the mixed material. Plural spray rig shall have heated hoppers, heated hoses to a mixer manifold through (at least two) static mixers to a 15-25 ft. 3/8" I.D. whip hose. Pre-mix the separate components prior to adding or incorporating into plural component equipment to break the gel. Do not heat material above 110°F (43°C). See "Phenoline Tank Shield Application Guide" for more detailed instructions.

APPLICATION EQUIPMENT GUIDELINES

Listed below are general equipment guidelines for the application of this product. Job site conditions may require modifications to these guidelines to achieve the desired results.

Plural Component Airless Spray

Equipment: 1:1 fixed-ratio plural component spray equipment with heated hoppers, in-line heaters, and heated hose bundles
Material Hose: 1/2" I.D. (min.)
Static Mixers: Two 1/2" x 12-element stainless steel
Whip Hose: 3/8" x 3-6 foot whip hose
Tip Size: 0.021"-0.027"
Optimal material temp: 100-105°F
See Carboline Application for complete instructions.

APPLICATION CONDITIONS

Condition	Material	Surface	Ambient	Humidity
Minimum	55°F (13°C)	35°F (2°C)	35°F (2°C)	0%
Maximum	90°F (32°C)	125°F (52°C)	110°F (43°C)	85%

This product requires the substrate temperature to be above the dew point. Condensation due to substrate temperatures below the dew point can cause flash rusting on prepared steel and interfere with proper adhesion to the substrate. Special application techniques may be required above or below normal application conditions.

CURING SCHEDULE

Surface Temp.	Dry to Handle	Immersion Service (Most Chemical Service)
35°F (2°C)	74 Hours	7 Days
50°F (10°C)	30 Hours	5 Days
75°F (24°C)	10 Hours	3 Days
90°F (32°C)	5 Hours	24 Hours

Dry to Touch and Dry to Recoat is normally 6 hours at 75°F (24°C).

Cure for Service: Cure for service times are dependent on curing conditions and expected immersion exposure. Film hardness (Shore D of 75 or greater) and/or solvent resistance (passes a 25 MEK solvent double-rubs*); are good indications that the lining is suitable for immersion service. Typically this can be from 24-72 hours or longer depending on the curing conditions.

Maximum recoat time is 30 days at 75°F (24°C) and reduces in half for each additional 15°F increase of surface temperature. If the product has exceeded the maximum recoat time, de-gloss and roughen by light sanding or mechanically abrade the surface and remove dust prior to topcoating.

*No significant color pick-up and some down-glossing is acceptable

CLEANUP & SAFETY

Cleanup | Thinner #2 or #76 are recommended for clean up.

Safety | Read and follow all caution statements on this product data sheet and on the SDS for this product.
Employ normal workmanlike safety precautions.

PACKAGING, HANDLING & STORAGE

Packaging

100 gal. kit (378 L)
10 Gal (37.8 L) Kit
4 Gal (15.1 L) Kit

Phenoline[®] Tank Shield

PRODUCT DATA SHEET



PACKAGING, HANDLING & STORAGE

Shelf Life	12 months
Storage Temperature & Humidity	40-110 °F (4-43 °C) 0-90% Relative Humidity
Storage	Store indoors
Shipping Weight (Approximate)	12 lbs/gal (5.5 kg/gal)
Flash Point (Setaflash)	Part A: 166 °F (74 °C) Part B: 204 °F (95 °C)

WARRANTY

To the best of our knowledge the technical data contained herein is true and accurate on the date of publication and is subject to change without prior notice. User must contact Carboline Company to verify correctness before specifying or ordering. No guarantee of accuracy is given or implied. We guarantee our products to conform to Carboline quality control. We assume no responsibility for coverage, performance, injuries or damages resulting from use. Carbolines sole obligation, if any, is to replace or refund the purchase price of the Carboline product(s) proven to be defective, at Carbolines option. Carboline shall not be liable for any loss or damage. NO OTHER WARRANTY OR GUARANTEE OF ANY KIND IS MADE BY CARBOLINE, EXPRESS OR IMPLIED, STATUTORY, BY OPERATION OF LAW, OR OTHERWISE, INCLUDING MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. All of the trademarks referenced above are the property of Carboline International Corporation unless otherwise indicated.

Selection & Specification Data

Generic Type Aliphatic Acrylic Polyurethane

- Features**
- High gloss
 - Excellent weatherability
 - Exceeds SSPC Paint 36 specification for a Level 3 urethane
 - Available in a variety of custom colors
 - Excellent flow characteristics allow for application by brush, spray, or roller
 - Superior impact and abrasion resistance
 - Indefinite recoatability

Description A high performance, high gloss polyurethane finish with excellent color and gloss retention properties. All finishes are minimum 80 gloss with exceptional weathering performance and are available in custom colors.

Performance Driven: Carbothane 134 series products represent a globally available product line that shares the same weathering performance criteria regardless of manufacturing location. Since raw materials are not always available (or compliant) on a global basis, these products have been uniquely formulated to utilize materials that are both available regionally and regulatory-compliant for the various regions around the world. All are formulated to provide the same weathering performance. The products have been rigorously tested and field proven giving specifiers the confidence that when specifying "Carbothane 134 Series" no matter where the project is, the product used will perform as expected.

Product Series The following products included in this series comply with the various regional or country regulatory requirements as indicated.

Product	SBV	VOC* (g/l)	Primary Area of Use
134 HG	70	264	North America
134 VOC	70	190	North America (CA)
134 MC	70	54	North America (CA-South Coast)
134 HP	65	300	Europe
134 GS	55	420	Outside North America
134 RS	46	478	Outside North America

*VOC as supplied

Dry Temp. Continuous: 93°C (200°F)
Resistance Non-Continuous: 121°C (250°F)
 Discoloration and loss of gloss is observed above 93°C (200°F).

Thickness and Coverage

Dry Film Thickness 50 µ (2.0 mils) nominal
 37-75 µ (1.5-3.0 mils) range as needed

Theoretical Coverage Rate at 1 mil (25 µ)

Product	m ² /liter	ft ² /gal
134 HG	27.54	1122
134 VOC	27.54	1122
134 MC	27.54	1122
134 HP	25.57	1042
134 GS	21.64	882
134 RS	18.11	738

Substrates & Surface Preparation

General Surfaces must be clean and dry. Employ adequate methods to remove dirt, dust, oil and all other contaminants that could interfere with adhesion of the coating. For all surfaces prime with specific Carboline primer as recommended by your Carboline sales representative. Refer to the specific primer's Product Data Sheet for detailed requirements of the specified primer.

Galvanized Steel Prime with specific Carboline primer as recommended by your Carboline Sales Representative. Refer to the specific primer's Product Data Sheet for substrate preparation requirements.

Previously Painted Surfaces Lightly sand or abrade to roughen and degloss the surface. Existing paint must attain a minimum 3B rating in accordance with ASTM D3359 "X-Cut" adhesion test.

Carbothane® 134 Product Series

Application Equipment

Listed below are general equipment guidelines for the application of this product. Job site conditions may require modification to these guidelines to achieve the desired results.

General Guidelines:

Spray Application (General) The following spray equipment has been found suitable and is available from equipment manufacturers.

Conventional Spray Pressure pot equipped with dual regulators, 3/8" I.D. minimum material hose, 0.070" I.D. fluid tip and appropriate air cap.

Airless Spray Pump Ratio: 30:1 (min.)*
GPM Output: 3.0 (min.)
Material Hose: 3/8" I.D. (min.)
Tip Size: .015-.017"
Output PSI: 2100-2400
Filter Size: 60 mesh
*PTFE packings are recommended and available from the pump manufacturer.

Brush & Roller (General) Multiple coats may be required to obtain desired appearance, recommended dry film thickness and adequate hiding. Avoid excessive re-brushing or re-rolling. For best results, tie-in within 10 minutes at 24°C (75°F).

Brush Recommended for touch-up only. Use a medium, natural bristle brush.

Roller Use a short-nap mohair roller cover with solvent resistant core.

Mixing & Thinning

Mixing Power mix Part A separately, then combine and power mix. DO NOT MIX PARTIAL KITS.

Thinning See separate product data sheet for specific product for appropriate thinning recommendations to comply with regulatory requirements and specific jobsite conditions. Use of thinners other than those supplied or recommended by Carboline may adversely affect product performance and void product warranty, whether expressed or implied.

Pot Life See specific product data sheet. Pot life ends when coating becomes too viscous to use. MOISTURE CONTAMINATION WILL SHORTEN POT LIFE AND CAUSE GELLATION.

Cleanup & Safety

Cleanup Use Thinner #2 or Acetone. In case of spillage, absorb and dispose of in accordance with local applicable regulations.

Safety Read and follow all caution statements on this product data sheet and on the MSDS for this product. Employ normal workmanlike safety precautions.

Ventilation When used in enclosed areas, thorough air circulation must be used during and after application until the coating is cured. The ventilation system should be capable of preventing the solvent vapor concentration from reaching the lower explosion limit for the solvents used. User should test and monitor exposure levels to insure all personnel are below guidelines. If not sure or if not able to monitor levels, use MSHA / NIOSH approved respirator.

Caution This product contains flammable solvents. Keep away from sparks and open flames. All electrical equipment and installations should be made and grounded in accordance with the National Electric Code. In areas where explosion hazards exist, workmen should be required to use non-ferrous tools and wear conductive and non-sparking shoes.

Application Conditions

Condition	Material	Surface	Ambient	Humidity
Minimum	10°C (50°F)	2°C (35°F)	2°C (35°F)	10%
Maximum	38°C (100°F)	49°C (120°F)	35°C (95°F)	85%

Industry standards are for substrate temperatures to be at least 3°C (5°F) above the dew point.

Caution: This product is moisture sensitive in the liquid stage and during initial cure. Protect from direct moisture (rain or dew) contact during initial curing period. Excessive exposure to moisture may result in a loss of gloss and/or microbubbling of the product.

Curing Schedule

See specific product data sheet for cure times. These times are based on a 2.0 mil (50 micron) dry film thickness. Higher film thickness, insufficient ventilation or cooler temperatures will require longer cure times and could result in solvent entrapment and premature failure.

*Maximum recoat times are indefinite. Surface must be clean and dry. As part of good painting practice it is recommended to test for adhesion by wiping the surface with solvent. If the film shows a slight "tack" the surface is suitable for recoating without extensive surface preparation such as abrading or de-glossing.

Handling & Storage

Storage (General) Store Indoors.

Storage Temperature & Humidity 4°-43°C (40° -110°F)
0-80% Relative Humidity


Coatings - Linings - Fireproofing
2150 Schuetz Rd., St. Louis, MO 63146
PH: 314-644-1000 Toll-Free: 800-848-4645
www.carboline.com

An  Company

September 2012 N

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SECTION - 5

Flyer



For General Dewatering Applications

Cost effective way to collect, dewater, transport and dump all kinds of sludges, slurries, and wet solids.

GritCat™ Model RO

The GritCat dewatering box comes with reusable filter media. The GritCat is designed to be used with gritty, or sandy type waste streams that typically do not require the use of a coagulant or flocculent for dewatering.

Log on to **ParkProcess.com** for more information.



Roll-Off

► Features

- Heavy duty carbon steel construction (optional stainless steel) 1/4" floor and 3/16" side walls with round bottom design, 6" x 2" x 1/4" bottom rails with steel wheels
- Gasketed, hinged water tight rear door
- Designed to be handled by a standard hook and cable type roll-off truck (hook lift type mechanisms optional)
- Typical (4) 3" drain ports located at the bottom on each side at opposing ends
- Filter media covers the floor, both side walls, and the front wall (optional door filter is available)
- Filter media comes in a choice of plastic or stainless steel and is held in place by bolts and brackets
- A double wall filter element that splits the container into two compartments is an available option (more than one double filter is also an option)
- Micron size filters from 3,360 microns to 170 microns (6 mesh to 80 mesh)
- Container sizes available: 15, 20, 25, 30, and 40 cubic yards



► Applications

- Waste Water Treatment Plants
- Vac Truck Waste
- Industrial Plant Sump Waste
- Truck & Heavy Equipment Wash Basin Waste

Dewatering Containers

- Self Dumping Hoppers
- Front Loaders
- Roll-Offs
- Permanently Mounted Tipping Stand
- Trailer Mounted
- Rotator Boxes
- Crane Dumping
- Filter Media

Filtration Equipment

- Cartridge Filter Vessels
- Cartridge Filters
- Bag Filter Vessels
- Bag Filters
- Filter Presses (Standard/Oilfield)
- Filter Press Cloths
- Sand Filters
- Back Wash Filters
- Carbon Filters
- Filter Media (Permanent /Disposable)

Separation Equipment

- Oil/Water Separators
- Hydrocyclones
- Vibratory Screening Units
- Inclined Plate Clarifier
- Grit Classifiers
- Strainers
- Oil Skimmers
- Mud Cleaners
- Centrifuges

Mixing/Blending

- Static Mixers
- Eductors
- Batch Mixing Systems
- Tank Mixers
- Agitators
- Blend Plants
- Blending Units
- Mud Shearing

Chemical Treatment

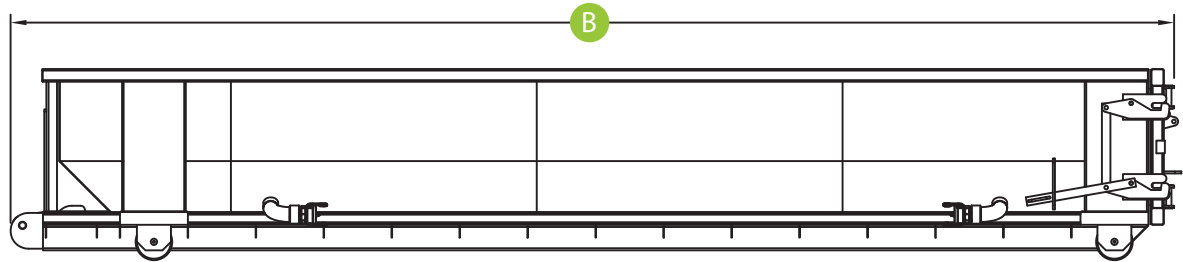
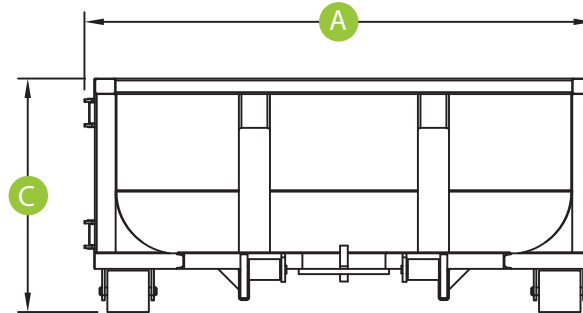
- Polymer Mixing & Injection
- Static Mixers
- PH Control Systems
- Slurry Tanks

Pumping

- Centrifugal Pumps
- Progressive Cavity Pumps
- Diaphragm Pumps
- Trash Pumps

Conveyance

- Screw Conveyors
- Drag Conveyors



Model	Size	A	B	C
GCRO15	15 yd	102	273.94	45
GCRO20	20 yd	102	273.94	55
GCRO25	25 yd	102	273.94	65

*Specifications subject to change without notice.
Measurements in inches and weights in pounds.

*NOTE: LARGER SIZES AVAILABLE UPON REQUEST

Other available models:



Self Dumping Hopper (DH)



Front Loader (FL)

SECTION - 6

Description/
Specification 2 yd
GritCat Dump Hopper

2 cu yd GritCat Dump Hopper

GRITCAT SELF-DUMPING HOPPER DEWATERING CONTAINER.

CONTAINER BODY IS MADE OF 7 GAUGE MILD STEEL WITH STRUCTURAL ANGLE FRAME AND STRUCTURAL BASE.

DESIGNED TO BE HANDLED BY A STANDARD FORKLIFT (BE SURE TO CHECK THE WEIGHT OF THE HOPPER AND LOAD TO MAKE SURE THE WEIGHT DOES NOT EXCEED THE CAPACITY OF THE FORKLIFT)

A FILTER SYSTEM IS INSTALLED INSIDE THE HOPPER COMPRISED OF FILTER SUPPORT PANELS BOLTED TO A WELDED IN FRAMEWORK.

ATTACHED TO THE TOP SURFACE OF THE SUPPORT PANELS IS A POLYESTER FILTER MEDIA, PPFM-62-350. WHICH IS HELD IN PLACE BY HOLD-DOWN FLAT BAR BRACKETS THAT GO AROUND THE PERIMETER OF EACH PANEL.

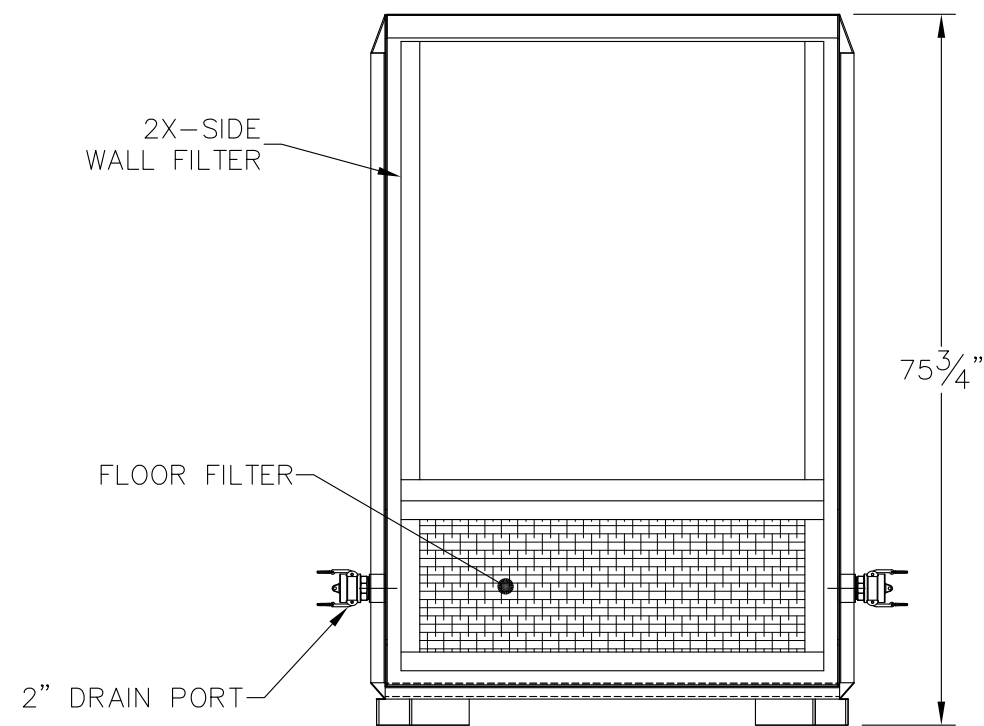
LOCATED ON THE BOTTOM BACK WALL (TYPICALLY) ARE LOCATED TWO 2" DRAINAGE PORTS WITH CAM LOCK HOSE FITTINGS AND CAM LOCK CAPS.

COATING:

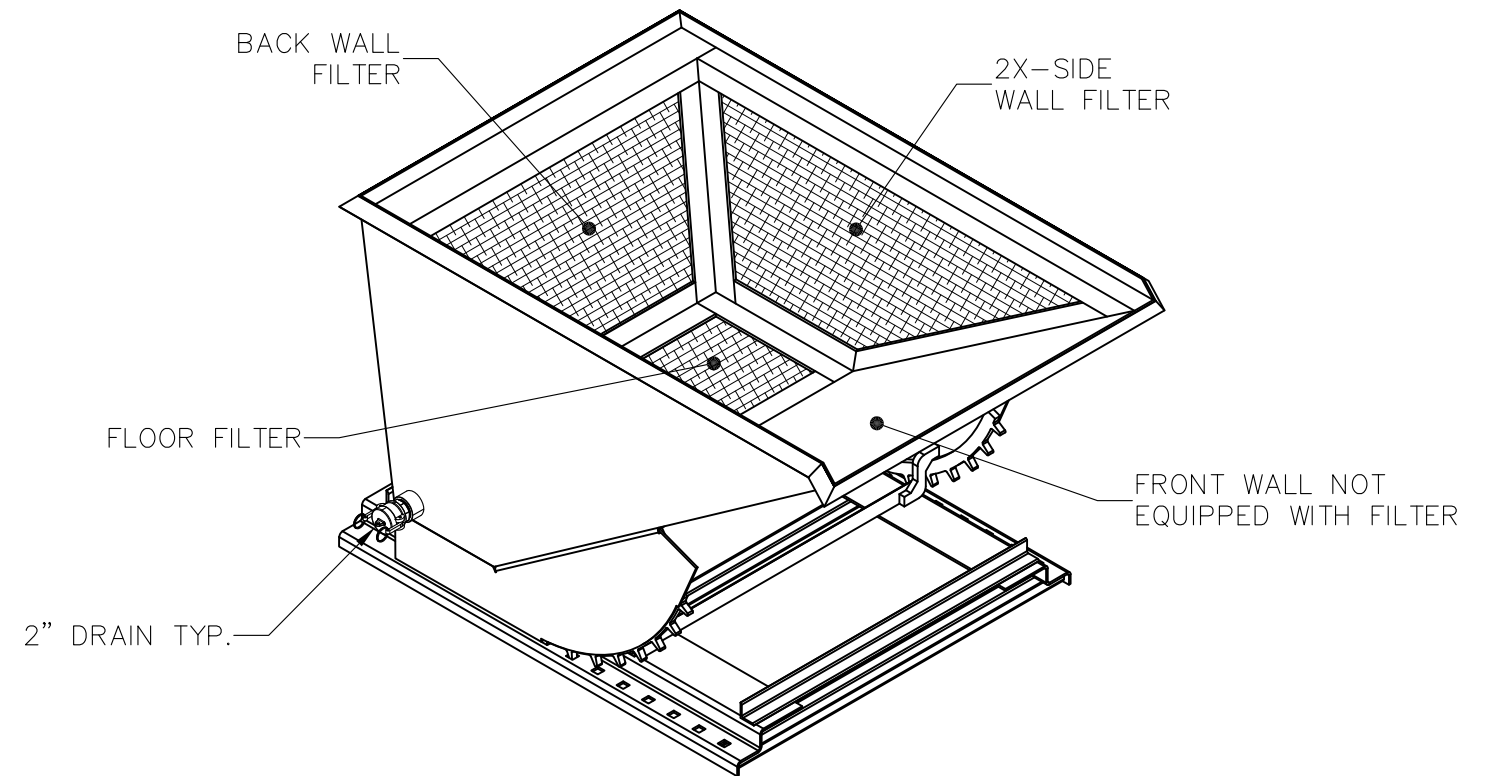
ALL CARBON STEEL SURFACES WILL BE SANDBLASTED WITH A SSPC-SP10 BLAST EXTERIOR IS COATED WITH A 2-PART CARBOLINE CARBOGUARD EPOXY TO 6 MILS AND UPPER COATED WITH CARBOTHANE POLYURETHANE PAINT TO 3 MILS. INTERIOR WILL BE COATED WITH 2 - 6 MIL THICK COATS OF CARBOGUARD. SUPPORT PANELS AND HOLD DOWN BRACKETS WILL BE HOT DIP GALVANIZED.

SECTION - 7

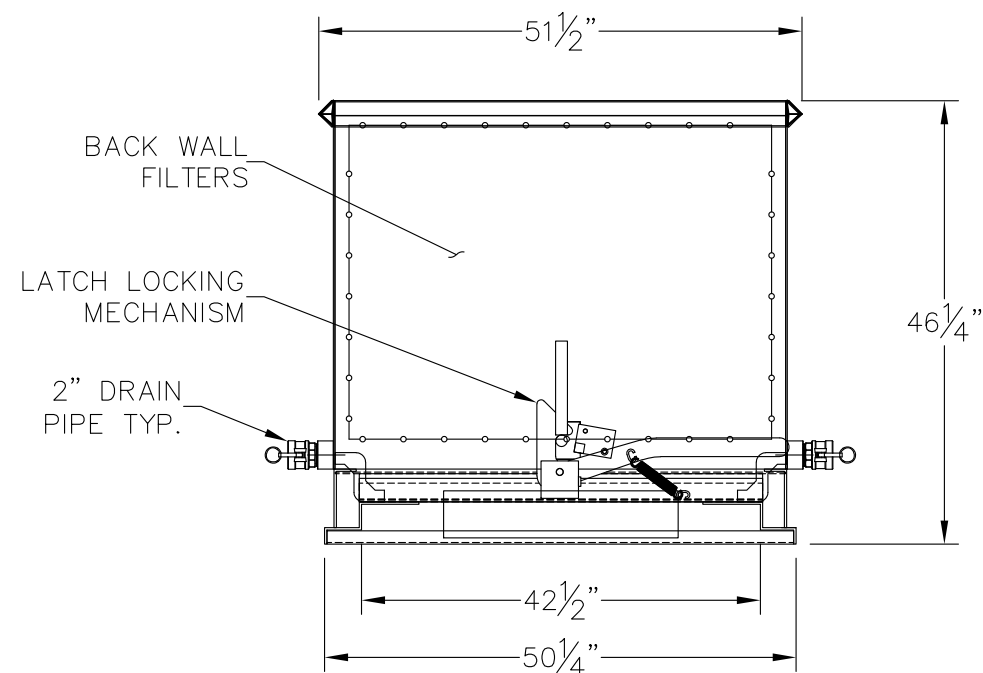
Drawings



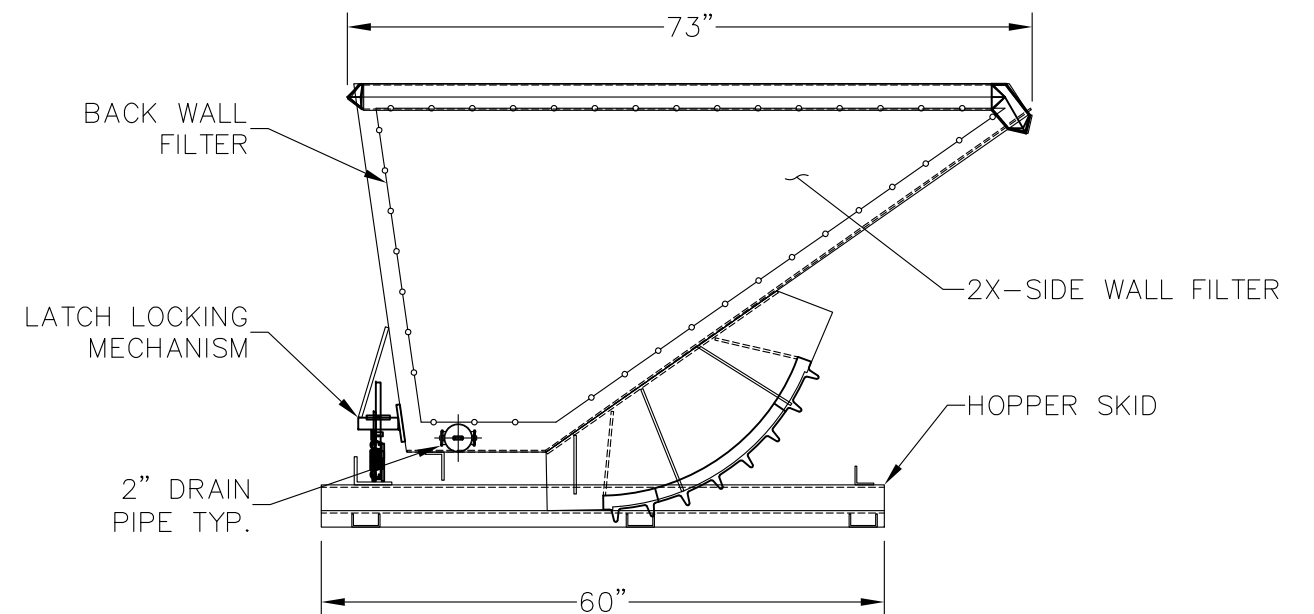
TOP VIEW
SOME ITEMS NOT SHOWN FOR CLARITY



ISOMETRIC VIEW



BACK VIEW
SOME ITEMS NOT SHOWN FOR CLARITY



SIDE VIEW

CUSTOMER APPROVAL:

SIGNED APPROVAL IS REQUIRED BEFORE BEGINNING CONSTRUCTION/FABRICATION.

PRINT NAME

SIGNATURE

DATE (MM/DD/YYYY)

NOTES:

1. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED OTHERWISE.
2. THIS STYLE IS EQUIPPED WITH A FILTER ON THE BOTTOM, BACK, LEFT, AND RIGHT SIDE(S) ONLY.

THIS SHEET IS THE PROPERTY OF PARK PROCESS, HOUSTON, TEXAS, AND MAY CONTAIN PROPRIETARY INFORMATION. IT IS CIRCULATED UNDER THE EXPRESS CONDITION THAT IT IS NOT TO BE USED IN ANY MANNER DIRECTLY OR INDIRECTLY DETRIMENTAL TO PARK PROCESS.

WEIGHTS, DIMENSIONS AND DESIGN ARE SUBJECT TO CHANGE. CONTACT PARK PROCESS FOR THE MOST UP TO DATE EQUIPMENT DRAWINGS AND INFORMATION.



2 YARD GRITCAT DUMP HOPPER-STYLE C
CUSTOMER APPROVAL

PM RC	DRN AU	CHK .	DWG. NO. PP-CA-GCDH-2C-01	REV. A
DATE	07/24/19			

SECTION - 8

Cut Sheets Sheets

SECTION - 8.A

Filter Media

GSM 62-350



Mesh (per inch)	68 X 20
Description (weave)	6/2 satin
Yarn Type:	Polyester / Polyester
Warp / Weft	.0197" / .0315" (500 microns / 800 microns)
Thickness	.085"
Weight	42.5 oz./yd ²
Air Permeability	325 - 355 CFM
Water Permeability	345 mm ³ / mm ² / sec
Tensile Strength	1560 lbs/inch
Micron Opening	0 x 325
Micron Retention	260
Open Area	19%
Available seams	316 SS clipper
Application:	WWT Belt Press



GSM ENGINEERED FABRICS

2015 Gateway Park Court
Kingsport, TN 37663

(423) 348-0954
www.gsmfiltration.com

SECTION - 8.B

Cam Lock Fittings



P-T Coupling Co., Inc.
 1414 E. Willow
 Enid,
 Phone: 580-237-4060
 Website: <http://www.ptcoupling.com>

F-Adapter (Adapter x Male NPT Thread)

PT maintains inventories at its 16 branch warehouse locations.

Couplings not interchangeable in the industry are sizes 5" and 8" through 12".

Made in the U.S.A

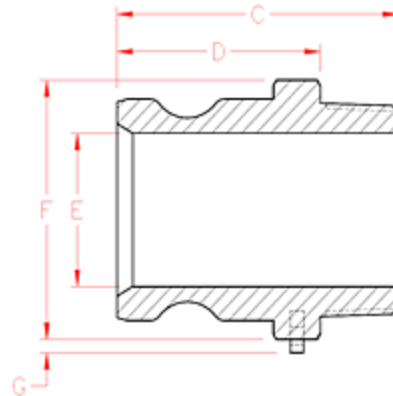
Note: Prices subject to change without notice.

More Information:

[Recommended Operating Conditions \(PDF\)](#)

[Safety Instructions & Recommended Materials \(PDF\)](#)

[PT Coupling Warranty Information \(PDF\)](#)



Results 1 - 99 of 99

<u>Item #</u>	<u>Part Name</u>	<u>Material</u>	<u>Adapter Size</u>	<u>Thread (Male NPT)</u>	<u>Inventory</u>	<u>List Price</u>
1000601	E05F	Aluminum	1/2 in.	1/2 in.	N/A	QUOTE
1200601	E05F	Brass	1/2 in.	1/2 in.	N/A	QUOTE
1400601	E05F	Stainless Steel	1/2 in.	1/2 in.	N/A	QUOTE
1800601	E05F	Ductile Iron	1/2 in.	1/2 in.	N/A	QUOTE
2000601	E05F	Hard Coat Aluminum	1/2 in.	1/2 in.	N/A	QUOTE
2400601	E05F	Carbon Steel	1/2 in.	1/2 in.	N/A	QUOTE
1000605	1/2F	Aluminum	3/4 in.	1/2 in.	N/A	QUOTE
1200605	1/2F	Brass	3/4 in.	1/2 in.	N/A	QUOTE
1400605	1/2F	Stainless Steel	3/4 in.	1/2 in.	N/A	QUOTE
1800605	1/2F	Ductile Iron	3/4 in.	1/2 in.	N/A	QUOTE
2000605	1/2F	Hard Coat Aluminum	3/4 in.	1/2 in.	N/A	QUOTE
2400605	1/2F	Carbon Steel	3/4 in.	1/2 in.	N/A	QUOTE
1000607	07F	Aluminum	3/4 in.	3/4 in.	N/A	QUOTE
1200607	07F	Brass	3/4 in.	3/4 in.	N/A	QUOTE

1400607	07F	Stainless Steel	3/4 in.	3/4 in.	N/A	QUOTE
1800607	07F	Ductile Iron	3/4 in.	3/4 in.	N/A	QUOTE
2000607	07F	Hard Coat Aluminum	3/4 in.	3/4 in.	N/A	QUOTE
2400607	07F	Carbon Steel	3/4 in.	3/4 in.	N/A	QUOTE
2700607	07F	Polypropylene	3/4 in.	3/4 in.	N/A	QUOTE
2770607	07F	Food Grade Poly	3/4 in.	3/4 in.	N/A	QUOTE
2900607	07F	Nylon	3/4 in.	3/4 in.	N/A	QUOTE
1000610	10F	Aluminum	1 in.	1 in.	N/A	QUOTE
1200610	10F	Brass	1 in.	1 in.	N/A	QUOTE
1400610	10F	Stainless Steel	1 in.	1 in.	N/A	QUOTE
1800610	10F	Ductile Iron	1 in.	1 in.	N/A	QUOTE
2000610	10F	Hard Coat Aluminum	1 in.	1 in.	N/A	QUOTE
2400610	10F	Carbon Steel	1 in.	1 in.	N/A	QUOTE
2700610	10F	Polypropylene	1 in.	1 in.	N/A	QUOTE
2770610	10F	Food Grade Poly	1 in.	1 in.	N/A	QUOTE
2900610	10F	Nylon	1 in.	1 in.	N/A	QUOTE
1000612	12F	Aluminum	1-1/4 in.	1-1/4 in.	N/A	QUOTE
1200612	12F	Brass	1-1/4 in.	1-1/4 in.	N/A	QUOTE
1400612	12F	Stainless Steel	1-1/4 in.	1-1/4 in.	N/A	QUOTE
1800612	12F	Ductile Iron	1-1/4 in.	1-1/4 in.	N/A	QUOTE
2000612	12F	Hard Coat Aluminum	1-1/4 in.	1-1/4 in.	N/A	QUOTE
2400612	12F	Carbon Steel	1-1/4 in.	1-1/4 in.	N/A	QUOTE
2700612	12F	Polypropylene	1-1/4 in.	1-1/4 in.	N/A	QUOTE
2770612	12F	Food Grade Poly	1-1/4 in.	1-1/4 in.	N/A	QUOTE
2900612	12F	Nylon	1-1/4 in.	1-1/4 in.	N/A	QUOTE
1000615	15F	Aluminum	1-1/2 in.	1-1/2 in.	N/A	QUOTE
1200615	15F	Brass	1-1/2 in.	1-1/2 in.	N/A	QUOTE
1400615	15F	Stainless Steel	1-1/2 in.	1-1/2 in.	N/A	QUOTE
1800615	15F	Ductile Iron	1-1/2 in.	1-1/2 in.	N/A	QUOTE
2000615	15F	Hard Coat Aluminum	1-1/2 in.	1-1/2 in.	N/A	QUOTE
2400615	15F	Carbon Steel	1-1/2 in.	1-1/2 in.	N/A	QUOTE
2700615	15F	Polypropylene	1-1/2 in.	1-1/2 in.	N/A	QUOTE
2770615	15F	Food Grade Poly	1-1/2 in.	1-1/2 in.	N/A	QUOTE
2900615	15F	Nylon	1-1/2 in.	1-1/2 in.	N/A	QUOTE
1000620	20F	Aluminum	2 in.	2 in.	N/A	QUOTE
1200620	20F	Brass	2 in.	2 in.	N/A	QUOTE
1400620	20F	Stainless Steel	2 in.	2 in.	N/A	QUOTE
1800620	20F	Ductile Iron	2 in.	2 in.	N/A	QUOTE
2000620	20F	Hard Coat Aluminum	2 in.	2 in.	N/A	QUOTE
2400620	20F	Carbon Steel	2 in.	2 in.	N/A	QUOTE
2700620	20F	Polypropylene	2 in.	2 in.	N/A	QUOTE
2770620	20F	Food Grade Poly	2 in.	2 in.	N/A	QUOTE
2900620	20F	Nylon	2 in.	2 in.	N/A	QUOTE
1000625	25F	Aluminum	2-1/2 in.	2-1/2 in.	N/A	QUOTE
1200625	25F	Brass	2-1/2 in.	2-1/2 in.	N/A	QUOTE

1400625	25F	Stainless Steel	2-1/2 in.	2-1/2 in.	N/A	QUOTE
1800625	25F	Ductile Iron	2-1/2 in.	2-1/2 in.	N/A	QUOTE
2000625	25F	Hard Coat Aluminum	2-1/2 in.	2-1/2 in.	N/A	QUOTE
2400625	25F	Carbon Steel	2-1/2 in.	2-1/2 in.	N/A	QUOTE
1000630	30F	Aluminum	3 in.	3 in.	N/A	QUOTE
1200630	30F	Brass	3 in.	3 in.	N/A	QUOTE
1400630	30F	Stainless Steel	3 in.	3 in.	N/A	QUOTE
1800630	30F	Ductile Iron	3 in.	3 in.	N/A	QUOTE
2000630	30F	Hard Coat Aluminum	3 in.	3 in.	N/A	QUOTE
2400630	30F	Carbon Steel	3 in.	3 in.	N/A	QUOTE
2700630	30F	Polypropylene	3 in.	3 in.	N/A	QUOTE
2770630	30F	Food Grade Poly	3 in.	3 in.	N/A	QUOTE
2900630	30F	Nylon	3 in.	3 in.	N/A	QUOTE
1000640	40F	Aluminum	4 in.	4 in.	N/A	QUOTE
1200640	40F	Brass	4 in.	4 in.	N/A	QUOTE
1400640	40F	Stainless Steel	4 in.	4 in.	N/A	QUOTE
1800640	40F	Ductile Iron	4 in.	4 in.	N/A	QUOTE
2000640	40F	Hard Coat Aluminum	4 in.	4 in.	N/A	QUOTE
2400640	40F	Carbon Steel	4 in.	4 in.	N/A	QUOTE
1000650	50F	Aluminum	5 in.	5 in.	N/A	QUOTE
1200650	50F	Brass	5 in.	5 in.	N/A	QUOTE
1400650	50F	Stainless Steel	5 in.	5 in.	N/A	QUOTE
1800650	50F	Ductile Iron	5 in.	5 in.	N/A	QUOTE
2000650	50F	Hard Coat Aluminum	5 in.	5 in.	N/A	QUOTE
1000660	60F	Aluminum	6 in.	6 in.	N/A	QUOTE
1200660	60F	Brass	6 in.	6 in.	N/A	QUOTE
1400660	60F	Stainless Steel	6 in.	6 in.	N/A	QUOTE
1800660	60F	Ductile Iron	6 in.	6 in.	N/A	QUOTE
2000660	60F	Hard Coat Aluminum	6 in.	6 in.	N/A	QUOTE
2400660	60F	Carbon Steel	6 in.	6 in.	N/A	QUOTE
1000680	80F	Aluminum	8 in.	8 in.	N/A	QUOTE
1200680	80F	Brass	8 in.	8 in.	N/A	QUOTE
1400680	80F	Stainless Steel	8 in.	8 in.	N/A	QUOTE
1800680	80F	Ductile Iron	8 in.	8 in.	N/A	QUOTE
2000680	80F	Hard Coat Aluminum	8 in.	8 in.	N/A	QUOTE
1000690	100F	Aluminum	10 in.	10 in.	N/A	QUOTE
1200690	100F	Brass	10 in.	10 in.	N/A	QUOTE
1400690	100F	Stainless Steel	10 in.	10 in.	N/A	QUOTE
1800690	100F	Ductile Iron	10 in.	10 in.	N/A	QUOTE
1000692	120F	Aluminum	12 in.	12 in.	N/A	QUOTE

Results 1 - 99 of 99



P-T Coupling Co., Inc.
 1414 E. Willow
 Enid,
Phone: 580-237-4060
Website: <http://www.ptcoupling.com>

V-Coupler (Dust Cap)

PT maintains inventories at its 16 branch warehouse locations.

HBS cam arms are made of 300 series Stainless Steel.

HB cam arms are made of Brass.

All Stainless Steel castings are 316 and all components are 300 series Stainless Steel.

Please call for special quote on complete cam arm assembly and components in 316 Stainless Steel.

8, 10 and 12 inch couplers are 4-cam designs with the STA-LOK® positive locking feature.

Couplings not interchangeable in the industry are sizes 5" and 8" through 12".

Made in the U.S.A.

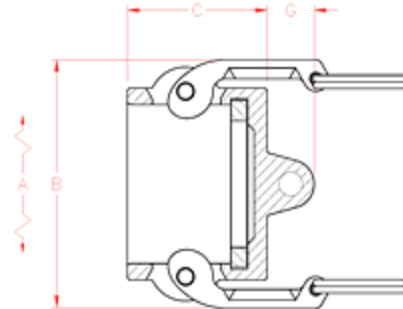
Note: Prices subject to change without notice.

More Information:

[Recommended Operating Conditions \(PDF\)](#)

[Safety Instructions & Recommended Materials \(PDF\)](#)

[PT Coupling Warranty Information \(PDF\)](#)



Results 1 - 119 of 119

<u>Item #</u>	<u>Part Name</u>	<u>Material</u>	<u>Coupler Size</u>	<u>Outside Diameter (B)</u>	<u>Inventory</u>	<u>List Price</u>
1001101	E05V	Aluminum/HBS	1/2 in.	1.82 in. 46 mm	N/A	QUOTE
1071101	E05V	Aluminum/HB	1/2 in.	1.82 in. 46 mm	N/A	QUOTE
1201101	E05V	Brass/HB	1/2 in.	1.82 in. 46 mm	N/A	QUOTE
1291101	E05V	Brass/HBS	1/2 in.	1.82 in. 46 mm	N/A	QUOTE

1401101	E05V	Stainless Steel	1/2 in.	1.82 in. 46 mm	N/A	QUOTE
1801101	E05V	Ductile Iron	1/2 in.	1.82 in. 46 mm	N/A	QUOTE
2001101	E05V	Hard Coat Aluminum	1/2 in.	1.82 in. 46 mm	N/A	QUOTE
1001107	07V	Aluminum/HBS	3/4 in.	2.11 in. 54 mm	N/A	QUOTE
1071107	07V	Aluminum/HB	3/4 in.	2.11 in. 54 mm	N/A	QUOTE
1201107	07V	Brass/HB	3/4 in.	2.11 in. 54 mm	N/A	QUOTE
1291107	07V	Brass/HBS	3/4 in.	2.11 in. 54 mm	N/A	QUOTE
1401107	07V	Stainless Steel	3/4 in.	2.11 in. 54 mm	N/A	QUOTE
1801107	07V	Ductile Iron	3/4 in.	2.11 in. 54 mm	N/A	QUOTE
2001107	07V	Hard Coat Aluminum	3/4 in.	2.11 in. 54 mm	N/A	QUOTE
2401107	07V	Carbon Steel	3/4 in.	2.11 in. 54 mm	N/A	QUOTE
2701107	07V	Polypropylene	3/4 in.	2.11 in. 54 mm	N/A	QUOTE
2771107	07V	Food Grade Poly	3/4 in.	2.11 in. 54 mm	N/A	QUOTE
2901107	07V	Nylon	3/4 in.	2.11 in. 54 mm	N/A	QUOTE
1001110	10V	Aluminum/HBS	1 in.	2.44 in. 62 mm	N/A	QUOTE
1071110	10V	Aluminum/HB	1 in.	2.44 in. 62 mm	N/A	QUOTE
1201110	10V	Brass/HB	1 in.	2.44 in. 62 mm	N/A	QUOTE
1291110	10V	Brass/HBS	1 in.	2.44 in. 62 mm	N/A	QUOTE
1401110	10V	Stainless Steel	1 in.	2.44 in. 62 mm	N/A	QUOTE
1801110	10V	Ductile Iron	1 in.	2.44 in. 62 mm	N/A	QUOTE
2001110	10V	Hard Coat Aluminum	1 in.	2.44 in. 62 mm	N/A	QUOTE
2401110	10V	Carbon Steel	1 in.	2.44 in. 62 mm	N/A	QUOTE
2701110	10V	Polypropylene	1 in.	2.44 in. 62 mm	N/A	QUOTE
2771110	10V	Food Grade Poly	1 in.	2.44 in. 62 mm	N/A	QUOTE
2901110	10V	Nylon	1 in.	2.44 in. 62 mm	N/A	QUOTE
1001112	12V	Aluminum/HBS	1-1/4 in.	3.26 in. 83 mm	N/A	QUOTE
1071112	12V	Aluminum/HB	1-1/4 in.	3.26 in. 83 mm	N/A	QUOTE

1201112	12V	Brass/HB	1-1/4 in.	3.26 in. 83 mm	N/A	QUOTE
1291112	12V	Brass/HBS	1-1/4 in.	3.26 in. 83 mm	N/A	QUOTE
1401112	12V	Stainless Steel	1-1/4 in.	3.26 in. 83 mm	N/A	QUOTE
1801112	12V	Ductile Iron	1-1/4 in.	3.26 in. 83 mm	N/A	QUOTE
2001112	12V	Hard Coat Aluminum	1-1/4 in.	3.26 in. 83 mm	N/A	QUOTE
2701112	12V	Polypropylene	1-1/4 in.	3.26 in. 83 mm	N/A	QUOTE
2771112	12V	Food Grade Poly	1-1/4 in.	3.26 in. 83 mm	N/A	QUOTE
1001115	15V	Aluminum/HBS	1-1/2 in.	3.56 in. 90 mm	N/A	QUOTE
1071115	15V	Aluminum/HB	1-1/2 in.	3.56 in. 90 mm	N/A	QUOTE
1201115	15V	Brass/HB	1-1/2 in.	3.56 in. 90 mm	N/A	QUOTE
1291115	15V	Brass/HBS	1-1/2 in.	3.56 in. 90 mm	N/A	QUOTE
1401115	15V	Stainless Steel	1-1/2 in.	3.56 in. 90 mm	N/A	QUOTE
1801115	15V	Ductile Iron	1-1/2 in.	3.56 in. 90 mm	N/A	QUOTE
2001115	15V	Hard Coat Aluminum	1-1/2 in.	3.56 in. 90 mm	N/A	QUOTE
2401115	15V	Carbon Steel	1-1/2 in.	3.56 in. 90 mm	N/A	QUOTE
2701115	15V	Polypropylene	1-1/2 in.	3.56 in. 90 mm	N/A	QUOTE
2771115	15V	Food Grade Poly	1-1/2 in.	3.56 in. 90 mm	N/A	QUOTE
2901115	15V	Nylon	1-1/2 in.	3.56 in. 90 mm	N/A	QUOTE
1001120	20V	Aluminum/HBS	2 in.	3.94 in. 100 mm	N/A	QUOTE
1071120	20V	Aluminum/HB	2 in.	3.94 in. 100 mm	N/A	QUOTE
1201120	20V	Brass/HB	2 in.	3.94 in. 100 mm	N/A	QUOTE
1291120	20V	Brass/HBS	2 in.	3.94 in. 100 mm	N/A	QUOTE
1401120	20V	Stainless Steel	2 in.	3.94 in. 100 mm	N/A	QUOTE
1801120	20V	Ductile Iron	2 in.	3.94 in. 100 mm	N/A	QUOTE
2001120	20V	Hard Coat Aluminum	2 in.	3.94 in. 100 mm	N/A	QUOTE
2401120	20V	Carbon Steel	2 in.	3.94 in. 100 mm	N/A	QUOTE
2701120	20V	Polypropylene	2 in.	3.94 in. 100 mm	N/A	QUOTE

2771120	20V	Food Grade Poly	2 in.	3.94 in. 100 mm	N/A	QUOTE
2901120	20V	Nylon	2 in.	3.94 in. 100 mm	N/A	QUOTE
1001125	25V	Aluminum/HBS	2-1/2 in.	4.43 in. 113 mm	N/A	QUOTE
1071125	25V	Aluminum/HB	2-1/2 in.	4.43 in. 113 mm	N/A	QUOTE
1201125	25V	Brass/HB	2-1/2 in.	4.43 in. 113 mm	N/A	QUOTE
1291125	25V	Brass/HBS	2-1/2 in.	4.43 in. 113 mm	N/A	QUOTE
1401125	25V	Stainless Steel	2-1/2 in.	4.43 in. 113 mm	N/A	QUOTE
1801125	25V	Ductile Iron	2-1/2 in.	4.43 in. 113 mm	N/A	QUOTE
2001125	25V	Hard Coat Aluminum	2-1/2 in.	4.43 in. 113 mm	N/A	QUOTE
2401125	25V	Carbon Steel	2-1/2 in.	4.43 in. 113 mm	N/A	QUOTE
1001130	30V	Aluminum/HBS	3 in.	5.46 in. 139 mm	N/A	QUOTE
1071130	30V	Aluminum/HB	3 in.	5.46 in. 139 mm	N/A	QUOTE
1201130	30V	Brass/HB	3 in.	5.46 in. 139 mm	N/A	QUOTE
1291130	30V	Brass/HBS	3 in.	5.46 in. 139 mm	N/A	QUOTE
1401130	30V	Stainless Steel	3 in.	5.46 in. 139 mm	N/A	QUOTE
1801130	30V	Ductile Iron	3 in.	5.46 in. 139 mm	N/A	QUOTE
2001130	30V	Hard Coat Aluminum	3 in.	5.46 in. 139 mm	N/A	QUOTE
2401130	30V	Carbon Steel	3 in.	5.46 in. 139 mm	N/A	QUOTE
2701130	30V	Polypropylene	3 in.	5.46 in. 139 mm	N/A	QUOTE
2771130	30V	Food Grade Poly	3 in.	5.46 in. 139 mm	N/A	QUOTE
2901130	30V	Nylon	3 in.	5.46 in. 139 mm	N/A	QUOTE
1001140	40V	Aluminum/HBS	4 in.	6.56 in. 167 mm	N/A	QUOTE
1071140	40V	Aluminum/HB	4 in.	6.56 in. 167 mm	N/A	QUOTE
1201140	40V	Brass/HB	4 in.	6.56 in. 167 mm	N/A	QUOTE
1291140	40V	Brass/HBS	4 in.	6.56 in. 167 mm	N/A	QUOTE
1401140	40V	Stainless Steel	4 in.	6.56 in. 167 mm	N/A	QUOTE
1801140	40V	Ductile Iron	4 in.	6.56 in. 167 mm	N/A	QUOTE

2001140	40V	Hard Coat Aluminum	4 in.	6.56 in. 167 mm	N/A	QUOTE
2401140	40V	Carbon Steel	4 in.	6.56 in. 167 mm	N/A	QUOTE
2701140	40V	Polypropylene	4 in.	6.56 in. 167 mm	N/A	QUOTE
2771140	40V	Food Grade Poly	4 in.	6.56 in. 167 mm	N/A	QUOTE
2901140	40V	Nylon	4 in.	6.56 in. 167 mm	N/A	QUOTE
1001150	50V	Aluminum/HBS	5 in.	7.57 in. 192 mm	N/A	QUOTE
1071150	50V	Aluminum/HB	5 in.	7.57 in. 192 mm	N/A	QUOTE
1201150	50V	Brass/HB	5 in.	7.57 in. 192 mm	N/A	QUOTE
1401150	50V	Stainless Steel	5 in.	7.57 in. 192 mm	N/A	QUOTE
1801150	50V	Ductile Iron	5 in.	7.57 in. 192 mm	N/A	QUOTE
2001150	50V	Hard Coat Aluminum	5 in.	7.57 in. 192 mm	N/A	QUOTE
1001160	60V	Aluminum/HBS	6 in.	10.16 in. 258 mm	N/A	QUOTE
1071160	60V	Aluminum/HB	6 in.	10.16 in. 258 mm	N/A	QUOTE
1201160	60V	Brass/HB	6 in.	10.16 in. 258 mm	N/A	QUOTE
1401160	60V	Stainless Steel	6 in.	10.16 in. 258 mm	N/A	QUOTE
1801160	60V	Ductile Iron	6 in.	10.16 in. 258 mm	N/A	QUOTE
2001160	60V	Hard Coat Aluminum	6 in.	10.16 in. 258 mm	N/A	QUOTE
2401160	60V	Carbon Steel	6 in.	10.16 in. 258 mm	N/A	QUOTE
1001180	80V	Aluminum/HBS	8 in.	12.13 in. 308 mm	N/A	QUOTE
1071180	80V	Aluminum/HB	8 in.	12.13 in. 308 mm	N/A	QUOTE
1201180	80V	Brass/HB	8 in.	12.13 in. 308 mm	N/A	QUOTE
1401180	80V	Stainless Steel	8 in.	12.13 in. 308 mm	N/A	QUOTE
1801180	80V	Ductile Iron	8 in.	12.13 in. 308 mm	N/A	QUOTE
2001180	80V	Hard Coat Aluminum	8 in.	12.13 in. 308 mm	N/A	QUOTE
2401180	80V	Carbon Steel	8 in.	12.13 in. 308 mm	N/A	QUOTE
1001190	100V	Aluminum/HBS	10 in.	14.56 in. 370 mm	N/A	QUOTE
1071190	100V	Aluminum/HB	10 in.	14.56 in. 370 mm	N/A	QUOTE

1201190	100V	Brass/HB	10 in.	14.56 in. 370 mm	N/A	QUOTE
1401190	100V	Stainless Steel	10 in.	14.56 in. 370 mm	N/A	QUOTE
1801190	100V	Ductile Iron	10 in.	14.56 in. 370 mm	N/A	QUOTE
1001192	120V	Aluminum/HBS	12 in.	16.62 in. 422 mm	N/A	QUOTE
1071192	120V	Aluminum/HB	12 in.	16.62 in. 422 mm	N/A	QUOTE
1401192	120V	Stainless Steel	12 in.	16.62 in. 422 mm	N/A	QUOTE
1801192	120V	Ductile Iron	12 in.	16.62 in. 422 mm	N/A	QUOTE

Results 1 - 119 of 119

SECTION - 8.C

Sponge Rubber Gasket



STYLE 759CCE

Style 759CCE is a closed cell Nitrile PVC medium density sponge gasket material. Excellent material for applications including gasket, seals, insulation, vibration and noise dampening.

Type	IV2
Polymer	NITRILE
Color	Black
ASTM D-1056 Classification	2B2
25% Compression Resistance (psi)	5.0 – 9.0
50% Compress Set (%)	40
Density (lb. /ft2)	5.5 – 7.5
Water Absorption (lb. /ft2)	0.1 max.
Tensile (psi)	75 min.
Elongation (%)	100 min.
Flammability – MVSS302	Pass
UL 94	Pass



250 Chester Street
Painesville, Ohio 44077
 Phone: **866-GO-AVERY (866-462-8379)**
 Fax: **888-358-4469**

email: psa.tape@averydennison.com

URL: stus.averydennison.com

Surface Preparation

It is essential, as with all pressure-sensitive tapes, that the surface to which the tape is applied be clean, dry, and free of grease or oil.

Temperatures

Min Application Temp	50° F	10° C
Max Continuous Operating Temp	150° F	66° C
Max Intermittent Operating Temp	175° F	79° C

Storage and Shelf Life

One year when stored at 70°F (21°C) 50% RH out of direct

Limited Warranty

All statements, technical information and recommendations concerning products sold or samples provided by AVERY DENNISON are based upon tests believed to be reliable but do not constitute a guarantee or warranty. All products are sold and samples of products provided with the understanding that PURCHASER has independently determined the suitability of such products for its purposes. AVERY DENNISON warrants the products to be free from defects in material and workmanship. Should any failure to conform to this warranty appear within one year* after the initial date of shipment, AVERY DENNISON shall, upon notification thereof and substantiation that the products have been stored and applied in accordance with AVERY DENNISON's standards, correct such defects by suitable repair or replacement without charge at AVERY DENNISON's plant or at the location of the products (at AVERY DENNISON's election) provided, however, if AVERY DENNISON determines that the repair or replacement is not commercially practical, AVERY DENNISON shall issue credit in favor of PURCHASER in an amount not to exceed the purchase price of the products.

This warranty is exclusive and is in lieu of any implied warranty of merchantability, fitness for a particular purpose or other warranty of quality, whether express or implied, except the warranty of title and against patent infringement. No waiver, alteration, additions or modifications of the foregoing conditions shall be valid unless made in writing and manually signed by an officer of Avery Dennison. *Or in the time period stated on the specific product specification sheet, if any, and if not then on the specific product information literature in effect at time of shipment.

Limitation of Liability

In no event shall AVERY DENNISON be liable for any incidental or consequential damages, including but not limited to, loss of profit, loss of use of production or loss of capital. The remedies of PURCHASER set forth herein are exclusive and the total liability of AVERY DENNISON with respect to any contract, or anything done in connection therewith such as the performance or breach hereof, or from the manufacture, sale, delivery, resale, installation or use of products whether arising out of contract, negligence, strict tort, or under any warranty, or otherwise, shall not exceed the purchase price of the products upon which the liability is based.

PRODUCT DESCRIPTION

Applications

Designed to provide an economical product for general purpose bonding.

Features

Polyester carrier
 Excellent tack and adhesion
 Densified kraft release liner

Benefits

Excellent processing and die-cutting stability
 Suited for a broad range of general purpose applications

PRODUCT DATA

Product Construction

Unwind Adhesive	Rubber Based
Carrier	Polyester
Liner Adhesive	Rubber Based
Liner	80 # Densified Kraft
Total Construction	

Test Methods: PSTC-133

<u>US Mil's</u>	<u>MM's</u>
2.0	0.05
0.5	0.01
1.5	0.04
4.6	0.12
8.6	0.22

Adhesive Properties

Peel Adhesion

2 mil PET 180° 12" / min Initial

<u>Substrate</u>	<u>Test Side</u>	<u>Lbf / in</u>	<u>US Oz / in</u>	<u>N / Meter</u>
SS	Liner	9.5	152	1665

Test Method(s): PSTC-103, ASTM D-3330, STD-10

ABS	Liner	8.3	132	1450
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PP	Liner	6.6	105	1152
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Talc PP	Liner	2.5	40	436
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LDPE	Liner	4.4	70	770
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HDPE	Liner	4.5	73	795
------	-------	------------	-----------	------------

Painted Metal	Liner	8.8	140	1537
---------------	-------	------------	------------	-------------

Static Shear

2 mil PET @
Substrate
 SS

Test Method(s): PSTC-107, ASTM D 3654, STD-9
 72°F / 22°C 1" sq (6.5 cm2) 5.5 lbs (2.5 kg)

<u>Test Side</u>	<u>Min to Fail</u>
Liner	> 1,200

SECTION - 8.D

Banjo Valve



POLYPROPYLENE VALVE: 2 INCH STANDARD PORT

PART NUMBER:
V200

TAB GROUP

- **Features and Benefits**
- **Technical**
 - Ball is diamond turned, after molding, to make it spherically perfect, which provides precise contact between ball and seats.
 - FKM (viton type) seals.
 - Precision molded polypropylene is reinforced with fiberglass for additional strength.
 - PTFE self-lubricating stem bushings and seats cannot stick or bind.
 - Self-aligning ball moves freely against the PTFE seats for smooth operation. The valve opens and closes with very little pressure on the handle.
 - Stainless steel bolts.

SECTION - 8.E

Fasteners



VIRAJ PROFILES LIMITED (FASTENER DIVISION)

TARAPUR, MAHARASHTRA, INDIA

MATERIAL TEST REPORT (CERTIFICATION TO EN-ISO 10204-3.1)

CUSTOMER : BRIGHTON-BEST INTERNATIONAL (TAIWAN) INC NO. 122,YILIN RD, RENDE TOWNSHIP TAIWAN COUNTY,717 TAIWAN ,R.O.C	MANUFACTURER'S		INVOICE NUMBER / MTR NO.		52-980433/24		
	SYMBOL		DATE		05.11.2014		
	VIRAJ OR 		ITEM DESCRIPTION		Stainless Steel 316 (Inch) Hex Nuts		
			SIZE		3/8" - 16		
			TOTAL QUANTITY		96000 Pieces	PART NO	763036
			MATERIAL SPECIFICATION		ASTM F594	LOT NO	041114002492
			CONDITION		CW1		
			DIMENSIONAL SPECIFICATION		ASME B 18.2.2		

CHEMICAL ANALYSIS

Sr.No	VALUES		Heat No	%C	%Mn	%Si	%S	%P	%Cr	%Ni	%Mo	%N2	Conforms to ASTM F594 Alloy Group-2 SS3316
	Required Values	Min											
		Max											
1	Observed Values		34262	0.017	1.35	0.37	0.005	0.036	16.85	10.14	2.01	0.035	

DIMENSIONAL REPORT / MECHANICAL PROPERTIES

Sr.No	Characteristic	Unit	Instrument	Required		No. of Sample Inspected	Observed Values		
				Min	Max		1	2	Sample Determinant
1	Width Across Flats	mm	Micrometer	14.00	14.27	As per lot	14.06	14.26	Qualified
2	Total Thickness	mm	Micrometer	8.13	8.55	As per lot	8.23	8.48	Qualified
3	Minor Diameter	mm	Micrometer	7.80	8.15	As per lot	7.85	7.96	Qualified
4	Flange Or Collar Diameter	mm	Micrometer	##NA	##NA	As per lot	##NA	##NA	Qualified
5	Gauge Fit / Thread Specification		Gauge	As per ASME B1.1		100 Nos	2B Go Answering freely. 2B No Go Not answering		
6	Proof Load	N	Tensile Testing M/c	34472		2 Nos	34558	34589	Qualified
7	Hardness	HRB	Hardness Tester	95HRB	106HRB	3 Nos	102	101	101
8	Marking	Visual		Logo Or VIRAJ	F594G 316	100 Nos	OK	OK	OK
9	Surface Discontinuities	Visual		As per ASTM F812-2002		100 Nos	No surface discontinuities observed		
10	Appearance	Visual				100 Nos	Satisfactory		
11	Surface Treatment		Passivation				Chemical Passivated and Conforms to ASTM A967-05(practice-A)		

Remarks :

DIMENSIONAL INSPECTION : CONFIRM TO THE SPECIFICATION

SURFACE INSPECTION : CONFIRM TO THE SPECIFICATION

USED WIRE ROD : FROM CONCAST BILLET

NA = NOT APPLICABLE

WE CERTIFY THAT THE MATERIAL DESCRIBED ABOVE HAS BEEN TESTED AND COMPLIES WITH THE ABOVE MENTIONED SPECIFICATION AND IS OF INDIAN ORIGIN AND FREE FROM RADIOACTIVE CONTAMINATION

Authorised Signatory - QA

TONG HWEI ENTERPRISE CO., LTD.

NO.399, GANGYAN ROAD, GANGSHAN 820, KAOHSIUNG TAIWAN, R.O.C. TEL: 886-7-6222851 FAX: 886-7-6211234

CERT OF INSPECTION

Cert no.: THB3060063-329 Page 1 of 1
STAR STAINLESS SCREW CO.
30 WEST END ROAD
TOTOWA NJ 07511
U. S. A.

Item no.: 32-FS-25C-125
Po# no.: QQQ120320JUN
Inv. no.: THB3060063
Issue date: Jun 26, 2013
Insp. date: Apr 07, 2012

Lot no.: B2407037
Ship qty: 20,800 PCS
Lot qty: 48,472 PCS
Sampler: H.G. TSAI
Sample plan spec.: ANSI/ASME B18.18.2M-1987

Mfr.: TONG HWEI ENTERPRISE CO., LTD.
Address: SAME AS LAB.
Mfr. date: Mar 28, 2012

SS 316CU; SKT FLAT CSK HEAD C/S 82+/-2, UNC; FULL THREAD; PASSIVATED; MARK "316" ON THE TOP HEAD
1/4-20 X 1-1/4 UNC 3A

Characteristic	Requirements	Results	S/S	Pass	Rej	Specification	Test Method
Dimensional inspections (Measurement by millimeters)							
Head Diameter	12.20 - 13.48	12.29 - 12.41	8	8	0	ANSI/ASME B18.3-02	
Nominal Length	30.73 - 31.75	31.16 - 31.22	32	32	0	ANSI/ASME B18.3-02	
Head Height	MIN 4.09	4.32 - 4.33	8	8	0	ANSI/ASME B18.3-02	
Socket Width Across Flats	3.97 - 4.03	PASSED	8	8	0	ANSI/ASME B18.3-02	
Major Diameter	6.150 - 6.350	6.151 - 6.198	32	32	0	ANSI/ASME B18.3-02	
Key Engagement	MIN 2.82	3.10 - 3.12	8	8	0	ANSI/ASME B18.3-02	
Pitch Diameter	5.460 - 5.520	5.470 - 5.480	32	32	0	ANSI/ASME B18.3-02	
Protrusion	1.10 - 1.62	1.47 - 1.48	8	8	0	ANSI/ASME B18.3-02	
Visual Appearance		PASSED	100	100	0	ASTM F788-08	
Mechanical properties							
Core Hardness (HRBW)	80 - 100	94 - 96	8	8	0	CUSTOMER REQUIREMENT	ASTM F606-11a
Tensile Strength (KSI)	80MIN	109.4 - 110.1	4	4	0	CUSTOMER REQUIREMENT	ASTM F606-11a
Yield Strength(0.2% offset)							N/A

Chemical composition %(Acc. to Material Inspection Certificate)					Heat No.: 3S228		Material Source: TAIWAN									
C	Si	Mn	P	S	Ni	Cr	Mo	Cu	N							
X100	X100	X100	X1000	X1000	X100	X100	X100	X100	ppm							
1.4	30	83	26	0.2	1050	1651	205	305								
Remarks: * TWO INDENTATION ONLY																

Remark: * TWO INDENTATION ONLY INSTEAD OF FOUR ON CORE HARDNESS.

All hardness requirements are consistent with THE standards if hardness requirements not provided to specification above

1. This inspection certificate is issued according to EN 10204 3.1/DIN 50049 3.1.
2. This report must not be reproduced except in full and to the item(s) tested only.
3. Whenever testing discrepancies are detected or departures from documented policies and procedure occur, they will be shown in remark.

329 of 372

Y. L. Su

Y. L. SU (QA ENGINEER)

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TONG HWEI ENTERPRISE CO., LTD.
NO.399, GANGYAN ROAD, GANGSHAN 820, KAOHSIUNG TAIWAN, R.O.C. TEL: 886-7-6222851 FAX: 886-7-6211234

CERT OF INSPECTION

Cert no.: THB5050018-834 Page 1 of 1
STAR STAINLESS SCREW CO.
30 WEST END ROAD
TOTOWA NJ 07511
U. S. A.

Item no.: 32-FS-25C-150
Po# no.: QQQ132175MAY
Inv. no.: B5050018
Issue date: May 25, 2015
Insp. date: Dec 09, 2013

Lot no.: B3C09083
Ship qty: 16,800 PCS
Lot qty: 27,029 PCS
Sampler: H.C. JIN

Mfr.: TONG HWEI ENTERPRISE CO., LTD.
Address: SAME AS LAB.
Mfr. date: Nov 25, 2013

SS 316CU; SKT FLAT CSK HEAD C/S 82+/-2, UNRC; FULL THREAD; PASSIVATED; MARK "316" ON THE TOP HEAD
1/4-20 X 1-1/2

Characteristic		Requirements	Results	S/S	Pass	Rej	Specification	Test Method / [Device*1] / (Sample Plan*2)						
Dimensional inspections														
Head Diameter (in)		0.480 - 0.531	0.4854 - 0.4858	15	15	0	ASME B18.3-2003 ASME B18.3-2003 ASME B18.3-2003 ASME B18.3-2003 ASME B18.3-2003 ASME B18.3-2003 ASME B1.1-2003 ASME B1.1-2003 ASTM F788-13	[Cal] (18)						
Socket Width Across Flats (in)		0.156 - 0.159	0.1567 - 0.1571	4	4	0		[Cal] (18)						
Gauge of hexagon sockets		5/32	PASSED	15	15	0		[Hex] (18)						
Key Engagement (in)		MIN 0.111	0.1189 - 0.1201	15	15	0		[Hep] (18)						
Protrusion (in)		0.043 - 0.064	0.0630 - 0.0630	15	15	0		[Hhg] (18)						
Nominal Length (in)		1.4600 - 1.5000	1.4969 - 1.4980	15	15	0		[Cal] (18)						
Major Diameter (in)		0.2419 - 0.2500	0.24646 - 0.24665	15	15	0		[Mcr] (18)						
Pitch Diameter (in)		0.2147 - 0.2175	0.21535 - 0.21575	15	15	0		[Tri] (18)						
Visual Appearance			PASSED	29	29	0		(18)						
Mechanical properties														
Core Hardness*3 (HRBW)		MIN 50 HRA	96 - 97	15	15	0	ASTM F879-12 ASTM F879-12	ASTM F606-14 [Rht] (14) ASTM F606-14 [Mtm] (14)						
Tensile Strength (KSI)		MIN 80.0	122.4 - 123.1	4	4	0								
Chemical composition % (Acc. to Material Inspection Certificate) Heat No.: 4E530														
C	Si	Mn	P	S	Ni	Cr	Mo	Cu	N	Material Source: TAIWAN				
X100	X100	X100	X1000	X1000	X100	X100	X100	X100	X10000					
4.2	32	120	28	2.7	1001	1610	204	36	272					
Remark:														
*1 Device: [Cal] Vernier caliper [Hep] Hexagon Penetration Gauge [Hhg] Hexagon Gauge [Mcr] Micrometer [Tri] Thread Gauge														

Remark:

- *1 Device: [Cal]Vernier caliper [Hep]Hexagon Penetration Gauge [Hex]Hexagon Plug gauge [Hhg]Protrusion-Head Height Gage [Mcr]Micrometer [Mtm]Material Testing Machine [Rht]Rockwell Hardness Tester [Tri]Tri-Roll Thread Comparator
*2 Sample Plan: (14)ASTM F1470-2012 MECHANICAL TEST (18)ASME B18.18-2011 CATEGORY 2
*3 Two indentation only

1. This inspection certificate is issued according to EN 10204 3.1/DIN 50049 3.1.
2. This report must not be reproduced except in full and to the item(s) tested only.
3. Whenever testing discrepancies are detected or departures from documented policies and procedure occur, they will be shown in remark.
4. If the products are needed to passivate, we confirm that all are passivated with slight liquid wax.
5. Approximate Rockwell Hardness Conversion Numbers From HRC or HRBW to HRA for Austenitic Stainless Steel, please refer to ASTM E140 Table 6.
834 of 918

Y. L. Su

Y. L. SU (QA ENGINEER)

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TONG HWEI ENTERPRISE CO., LTD.

NO.399, GANGYAN ROAD, GANGSHAN 820, KAOHSIUNG TAIWAN, R.O.C. TEL: 886-7-6222851 FAX: 886-7-6211234

CERT OF INSPECTION

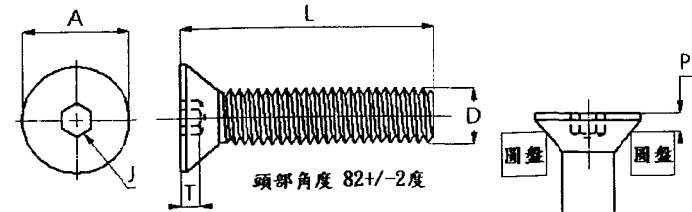
Cert no.: THB6040107-764 Page 1 of 1

STAR STAINLESS SCREW CO.
30 WEST END ROAD
TOTOWA NJ 07511
U. S. A.

SS 316CU; SKT FLAT CSK HEAD C/S 82+/-2, UNRC; FULL
THREAD; PASSIVATED; MARK "316" ON THE TOP HEAD

3/8-16 X 1-1/4

Lot no.: B6307184
Item no.: 32-FS-37C-125
Po# no.: QQQ138409MAY
Inv. no.: B6040107
Issue date: Apr 29, 2016
Insp. date: Mar 07, 2016
Ship qty: 20,000 PCS
Lot qty: 26,082 PCS
Sampler: H.G. TSAI
Mfr/Addr: Same as Lab
Mfr. date: Mar 04, 2016



Characteristic	Requirements	Results	S/S	Pass	Rej	Specification*1	Test Method [Device*2] (Sample Plan*3)
Dimensional inspections							
A Head Diameter (in)	0.720 - 0.781	0.7205 - 0.7280	15	15	0	ASME B18.3-2012	[Cal] (18)
J Socket Width Across Flats (in)	0.219 - 0.222	0.2209 - 0.2213	4	4	0	ASME B18.3-2012	[Cal] (18)
GJ Gauge of hexagon sockets	7/32	PASSED	15	15	0	ASME B18.3-2012	[Hex] (18)
T Key Engagement (in)	MIN 0.159	0.1732 - 0.1748	15	15	0	ASME B18.3-2012	[Hep] (18)
P Protrusion (in)	0.050 - 0.088	0.0756 - 0.0760	15	15	0	ASME B18.3-2012	[Hhg] (18)
L Nominal Length (in)	1.2100 - 1.2500	1.2331 - 1.2343	15	15	0	ASME B18.3-2012	[Cal] (18)
D Major Diameter (in)	0.3656 - 0.3750	0.37177 - 0.37209	15	15	0	ASME B1.1-2003	[Mcr] (18)
PD Functional Dia.	UNRC 3A	PASSED	15	15	0	ASME B1.1-2003	[Trg] (18)
PD Pitch Diameter (in)	0.331 - 0.334	0.33110 - 0.33157	15	15	0	ASME B1.1-2003	[Tri] (18)
VA Visual Appearance		PASSED	29	29	0	ASTM F788-13	(18)
Mechanical properties							
HR Core Hardness*4 (HRBW)	MIN 50 HRA	98 - 100	15	15	0	ASTM F879-12	ASTM F606-14 [Rht] (14)
TS Tensile Strength (KSI)	MIN 80.0	109.9 - 110.8	4	4	0	ASTM F879-12	ASTM F606-14 [Mtm] (14)
EL Extension (in)	MIN 0.150	0.1646 - 0.1732	4	4	0	ASTM F879-12	ASTM F606-14 [Cal] (14)

Chemical composition % (Acc. to Material Inspection Certificate)										Heat No.: 4T430					Material Source: TAIWAN				
C	Si	Mn	P	S	Ni	Cr	Mo	Cu	N										
X100	X100	X100	X1000	X1000	X100	X100	X100	X100	X10000										
1.9	29	80	27	0.8	1053	1662	203	304	241										

Remark:

*1 Additional explain: N/A

*2 Device: [Cal] Vernier caliper [Hep] Hexagon Penetration Gauge [Hex] Hexagon Plug gauge [Hhg] Protrusion-Head Height Gage [Mcr] Micrometer [Mtm] Material Testing Machine [Rht] Rockwell Hardness

Tester [Trg] Thread Ring Gauge [Tri] Tri-Roll Thread Comparator

*3 Sample Plan: (14) ASTM F1470-2012 MECHANICAL TEST (18) ASME B18.18-2011 CATEGORY 2

*4 Three indentation only

- This inspection certificate is issued according to EN 10204 3.1/DIN 50049 3.1. It must not be reproduced except in full and to the item(s) tested only.
- Product meets the order requirements.
- Whenever testing discrepancies are detected or departures from documented policies and procedure occur, they will be shown in remark.
- If the products are needed to passivate, we confirm that all are passivated with slight liquid wax. Passivation test per ASTM A380-2006: Passed.
- Approximate Rockwell Hardness Conversion Numbers From HRC or HRBW to HRA for Austenitic Stainless Steel, please refer to ASTM E140 Table 6.

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SECTION - 9

Paint Specification

SELECTION & SPECIFICATION DATA

Generic Type	Phenolic epoxy novolac
Description	This product is a solvent-free, high performance epoxy coating designed as an internal tank, valve and pipe lining for chemical or other commodity storage. It is a unique blend of resins and curing agents that allow batch mixing for ease of application. Plural component spray equipment is not required. The product is blush resistant and is typically applied at film thicknesses of 20 mils (500 microns) or thicker as needed (tank floors). It can handle exposures typically seen in the oil and gas industries; crude oils and fuels. It is resistant to NGL condensates, produced water, brines, industrial process water, waste water, and sewage. Ideal for municipal wastewater and water treatment facilities. Phenoline Tank Shield may be used as a patching and seam sealer by adding Carboline Thixatropo D to the pre-mixed kit at a ratio between 2:1 and 1:1 by volume resin:powder.
Features	• Batch mix formulation, single leg airless spray • High impact resistance • Superior adhesion to steel • Excellent resistance to water and salt water • Resistance to a broad range of fuels • Passes EI 1541 requirements for internal protective coating systems used in aviation fuel handling systems, including the jet fuel gum test • Adheres to API 653 inspection and API 652 guidelines for inspection intervals. • Resistant to hot water up to 180°F (82°C) • Excellent abrasion resistance and flexibility • Can be applied down to 35 °F (2 °C) • Can be applied as a single or multi-coat system • Non-blushing with a long recoat window • Low odor
Color	Standard: Grey (N700), White (N800), Blue (N100)
Finish	Gloss
Primer	Coating is normally applied direct to metal. May be applied over other primers as recommended by Carboline.
Dry Film Thickness	12 - 30 mils (305 - 762 microns) per coat Depends on service and existing condition of the substrate, product is typically applied in a one coat application at the appropriate film thickness depending on the application. Higher film thicknesses (60+ mils/1500+ microns) are used for more aggressive or abrasive conditions or for severely pitted steel (tank bottoms). Maximum vertical film build is 30 mils (750 microns).
Solids Content	By Volume 99% +/- 1%
Theoretical Coverage Rate	1588 ft²/gal at 1.0 mils (39.0 m²/l at 25 microns) 132 ft²/gal at 12.0 mils (3.2 m²/l at 300 microns) 53 ft²/gal at 30.0 mils (1.3 m²/l at 750 microns) Allow for loss in mixing and application.
VOC Values	As Supplied : 9 g/l
Wet Temp. Resistance	Immersion temperature resistance depends upon exposure, consult Carboline Technical Service for specific information

Phenoline[®] Tank Shield

PRODUCT DATA SHEET



SUBSTRATES & SURFACE PREPARATION

General	Surfaces must be clean and dry. Employ adequate methods to remove dirt, dust, oil and all other contaminants that could interfere with adhesion of the coating
Steel	Cleanliness: Abrasive blast to SSPC-SP10 (minimum) Profile: Minimum 3 mil (75 micron) dense, sharp anchor profile free of peening, as measured by ASTM D 4417. Defects exposed by blasting must be repaired.
Stainless Steel	Prepare by abrasive blasting to SSPC-SP 17 Thorough Abrasive Blast to a minimum of 3 mils (75 microns) dense angular anchor profile.
Concrete	Concrete: Clean and dry. Remove all loose, unsound concrete. Do not apply coating unless concrete has cured at least 28 days @ 70°F (21°C) and 50% RH or equivalent. Prepare surfaces in accordance with ASTM D4258 Surface Cleaning of Concrete and ASTM D4259 Abrading Concrete. Voids in concrete may require filling/surfacing.

MIXING & THINNING

Mixing	This product may be batch mixed and applied using standard airless spray equipment. IMPORTANT: Power mix each component separately, then combine and power mix until homogenous. Component Details for Colors: Grey (N700): The Part A is black (N909) and the Part B is white (N800) Blue (N100): The Part A is blue (N910) and the Part B is white (N800) White (N800): The Part A is clear (N000) and the Part B is white (N800)
Thinning	Thinning not normally required.
Ratio	1:1 by volume (Part A to Part B)
Pot Life	30 minutes at 75°F (24°C). Consult Carboline Technical Service for techniques to maximize pot life.

APPLICATION EQUIPMENT GUIDELINES

Listed below are general equipment guidelines for the application of this product. Job site conditions may require modifications to these guidelines to achieve the desired results.

General	This is a high solids coating and may require adjustments in spray techniques. Wet film thickness is easily and quickly achieved. The following spray equipment has been found suitable and is available from equipment manufacturers.
Airless Spray	Airless spray equipment capable of minimum 6000 psi (60:1 ratio or higher is preferred) with a minimum 180 cc lower. Fluid hose shall be minimum 3/8" I.D. Airless spray gun shall be rated minimum 7000 psi utilizing reverse-a-clean tips sizes 0.021-0.027" with fan size range between #5 to #9. A wider tip fan size facilitates break up and reduces fingering. Fixed-ratio (1:1 by volume) plural component equipment may also be used if the material cannot be sprayed within the pot life of the mixed material. Plural spray rig shall have heated hoppers, heated hoses to a mixer manifold through (at least two) static mixers to a 15-25 ft. 3/8" I.D. whip hose. Pre-mix the separate components prior to adding or incorporating into plural component equipment to break the gel. Do not heat material above 110°F (43°C). See "Phenoline Tank Shield Application Guide" for more detailed instructions.

APPLICATION EQUIPMENT GUIDELINES

Listed below are general equipment guidelines for the application of this product. Job site conditions may require modifications to these guidelines to achieve the desired results.

Plural Component Airless Spray

Equipment: 1:1 fixed-ratio plural component spray equipment with heated hoppers, in-line heaters, and heated hose bundles
Material Hose: 1/2" I.D. (min.)
Static Mixers: Two 1/2" x 12-element stainless steel
Whip Hose: 3/8" x 3-6 foot whip hose
Tip Size: 0.021"-0.027"
Optimal material temp: 100-105°F
See Carboline Application for complete instructions.

APPLICATION CONDITIONS

Condition	Material	Surface	Ambient	Humidity
Minimum	55°F (13°C)	35°F (2°C)	35°F (2°C)	0%
Maximum	90°F (32°C)	125°F (52°C)	110°F (43°C)	85%

This product requires the substrate temperature to be above the dew point. Condensation due to substrate temperatures below the dew point can cause flash rusting on prepared steel and interfere with proper adhesion to the substrate. Special application techniques may be required above or below normal application conditions.

CURING SCHEDULE

Surface Temp.	Dry to Handle	Immersion Service (Most Chemical Service)
35°F (2°C)	74 Hours	7 Days
50°F (10°C)	30 Hours	5 Days
75°F (24°C)	10 Hours	3 Days
90°F (32°C)	5 Hours	24 Hours

Dry to Touch and Dry to Recoat is normally 6 hours at 75°F (24°C).

Cure for Service: Cure for service times are dependent on curing conditions and expected immersion exposure. Film hardness (Shore D of 75 or greater) and/or solvent resistance (passes a 25 MEK solvent double-rubs*); are good indications that the lining is suitable for immersion service. Typically this can be from 24-72 hours or longer depending on the curing conditions.

Maximum recoat time is 30 days at 75°F (24°C) and reduces in half for each additional 15°F increase of surface temperature. If the product has exceeded the maximum recoat time, de-gloss and roughen by light sanding or mechanically abrade the surface and remove dust prior to topcoating.

*No significant color pick-up and some down-glossing is acceptable

CLEANUP & SAFETY

Cleanup | Thinner #2 or #76 are recommended for clean up.

Safety | Read and follow all caution statements on this product data sheet and on the SDS for this product. Employ normal workmanlike safety precautions.

PACKAGING, HANDLING & STORAGE

Packaging

100 gal. kit (378 L)
10 Gal (37.8 L) Kit
4 Gal (15.1 L) Kit

Phenoline[®] Tank Shield

PRODUCT DATA SHEET



PACKAGING, HANDLING & STORAGE

Shelf Life	12 months
Storage Temperature & Humidity	40-110 °F (4-43 °C) 0-90% Relative Humidity
Storage	Store indoors
Shipping Weight (Approximate)	12 lbs/gal (5.5 kg/gal)
Flash Point (Setaflash)	Part A: 166 °F (74 °C) Part B: 204 °F (95 °C)

WARRANTY

To the best of our knowledge the technical data contained herein is true and accurate on the date of publication and is subject to change without prior notice. User must contact Carboline Company to verify correctness before specifying or ordering. No guarantee of accuracy is given or implied. We guarantee our products to conform to Carboline quality control. We assume no responsibility for coverage, performance, injuries or damages resulting from use. Carbolines sole obligation, if any, is to replace or refund the purchase price of the Carboline product(s) proven to be defective, at Carbolines option. Carboline shall not be liable for any loss or damage. NO OTHER WARRANTY OR GUARANTEE OF ANY KIND IS MADE BY CARBOLINE, EXPRESS OR IMPLIED, STATUTORY, BY OPERATION OF LAW, OR OTHERWISE, INCLUDING MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. All of the trademarks referenced above are the property of Carboline International Corporation unless otherwise indicated.

Selection & Specification Data

Generic Type Aliphatic Acrylic Polyurethane

- Features**
- High gloss
 - Excellent weatherability
 - Exceeds SSPC Paint 36 specification for a Level 3 urethane
 - Available in a variety of custom colors
 - Excellent flow characteristics allow for application by brush, spray, or roller
 - Superior impact and abrasion resistance
 - Indefinite recoatability

Description A high performance, high gloss polyurethane finish with excellent color and gloss retention properties. All finishes are minimum 80 gloss with exceptional weathering performance and are available in custom colors.

Performance Driven: Carbothane 134 series products represent a globally available product line that shares the same weathering performance criteria regardless of manufacturing location. Since raw materials are not always available (or compliant) on a global basis, these products have been uniquely formulated to utilize materials that are both available regionally and regulatory-compliant for the various regions around the world. All are formulated to provide the same weathering performance. The products have been rigorously tested and field proven giving specifiers the confidence that when specifying "Carbothane 134 Series" no matter where the project is, the product used will perform as expected.

Product Series The following products included in this series comply with the various regional or country regulatory requirements as indicated.

Product	SBV	VOC* (g/l)	Primary Area of Use
134 HG	70	264	North America
134 VOC	70	190	North America (CA)
134 MC	70	54	North America (CA-South Coast)
134 HP	65	300	Europe
134 GS	55	420	Outside North America
134 RS	46	478	Outside North America

*VOC as supplied

Dry Temp. Continuous: 93°C (200°F)
Resistance Non-Continuous: 121°C (250°F)
 Discoloration and loss of gloss is observed above 93°C (200°F).

Thickness and Coverage

Dry Film Thickness 50 µ (2.0 mils) nominal
 37-75 µ (1.5-3.0 mils) range as needed

Theoretical Coverage Rate at 1 mil (25 µ)

Product	m ² /liter	ft ² /gal
134 HG	27.54	1122
134 VOC	27.54	1122
134 MC	27.54	1122
134 HP	25.57	1042
134 GS	21.64	882
134 RS	18.11	738

Substrates & Surface Preparation

General Surfaces must be clean and dry. Employ adequate methods to remove dirt, dust, oil and all other contaminants that could interfere with adhesion of the coating. For all surfaces prime with specific Carboline primer as recommended by your Carboline sales representative. Refer to the specific primer's Product Data Sheet for detailed requirements of the specified primer.

Galvanized Steel Prime with specific Carboline primer as recommended by your Carboline Sales Representative. Refer to the specific primer's Product Data Sheet for substrate preparation requirements.

Previously Painted Surfaces Lightly sand or abrade to roughen and degloss the surface. Existing paint must attain a minimum 3B rating in accordance with ASTM D3359 "X-Cut" adhesion test.

Application Equipment

Listed below are general equipment guidelines for the application of this product. Job site conditions may require modification to these guidelines to achieve the desired results.

General Guidelines:

Spray Application (General) The following spray equipment has been found suitable and is available from equipment manufacturers.

Conventional Spray Pressure pot equipped with dual regulators, 3/8" I.D. minimum material hose, 0.070" I.D. fluid tip and appropriate air cap.

Airless Spray Pump Ratio: 30:1 (min.)*
GPM Output: 3.0 (min.)
Material Hose: 3/8" I.D. (min.)
Tip Size: .015-.017"
Output PSI: 2100-2400
Filter Size: 60 mesh
*PTFE packings are recommended and available from the pump manufacturer.

Brush & Roller (General) Multiple coats may be required to obtain desired appearance, recommended dry film thickness and adequate hiding. Avoid excessive re-brushing or re-rolling. For best results, tie-in within 10 minutes at 24°C (75°F).

Brush Recommended for touch-up only. Use a medium, natural bristle brush.

Roller Use a short-nap mohair roller cover with solvent resistant core.

Mixing & Thinning

Mixing Power mix Part A separately, then combine and power mix. DO NOT MIX PARTIAL KITS.

Thinning See separate product data sheet for specific product for appropriate thinning recommendations to comply with regulatory requirements and specific jobsite conditions. Use of thinners other than those supplied or recommended by Carboline may adversely affect product performance and void product warranty, whether expressed or implied.

Pot Life See specific product data sheet. Pot life ends when coating becomes too viscous to use. MOISTURE CONTAMINATION WILL SHORTEN POT LIFE AND CAUSE GELLATION.

Cleanup & Safety

Cleanup Use Thinner #2 or Acetone. In case of spillage, absorb and dispose of in accordance with local applicable regulations.

Safety Read and follow all caution statements on this product data sheet and on the MSDS for this product. Employ normal workmanlike safety precautions.

Ventilation When used in enclosed areas, thorough air circulation must be used during and after application until the coating is cured. The ventilation system should be capable of preventing the solvent vapor concentration from reaching the lower explosion limit for the solvents used. User should test and monitor exposure levels to insure all personnel are below guidelines. If not sure or if not able to monitor levels, use MSHA / NIOSH approved respirator.

Caution This product contains flammable solvents. Keep away from sparks and open flames. All electrical equipment and installations should be made and grounded in accordance with the National Electric Code. In areas where explosion hazards exist, workmen should be required to use non-ferrous tools and wear conductive and non-sparking shoes.

Application Conditions

Condition	Material	Surface	Ambient	Humidity
Minimum	10°C (50°F)	2°C (35°F)	2°C (35°F)	10%
Maximum	38°C (100°F)	49°C (120°F)	35°C (95°F)	85%

Industry standards are for substrate temperatures to be at least 3°C (5°F) above the dew point.

Caution: This product is moisture sensitive in the liquid stage and during initial cure. Protect from direct moisture (rain or dew) contact during initial curing period. Excessive exposure to moisture may result in a loss of gloss and/or microbubbling of the product.

Curing Schedule

See specific product data sheet for cure times. These times are based on a 2.0 mil (50 micron) dry film thickness. Higher film thickness, insufficient ventilation or cooler temperatures will require longer cure times and could result in solvent entrapment and premature failure.

*Maximum recoat times are indefinite. Surface must be clean and dry. As part of good painting practice it is recommended to test for adhesion by wiping the surface with solvent. If the film shows a slight "tack" the surface is suitable for recoating without extensive surface preparation such as abrading or de-glossing.

Handling & Storage

Storage (General) Store Indoors.

Storage Temperature & Humidity 4°-43°C (40° -110°F)
0-80% Relative Humidity


Coatings - Linings - Fireproofing
2150 Schuetz Rd., St. Louis, MO 63146
PH: 314-644-1000 Toll-Free: 800-848-4645
www.carboline.com

An  Company

September 2012 N

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SECTION - 10

Flyers



For General Dewatering Applications

Cost effective way to collect, dewater, transport and dump all kinds of sludges, slurries, and wet solids.

GritCat™ Model DH

The GritCat dewatering box comes with reusable filter media. The GritCat is designed to be used with gritty or sandy type waste streams that typically do not require the use of a coagulant or flocculent for dewatering.

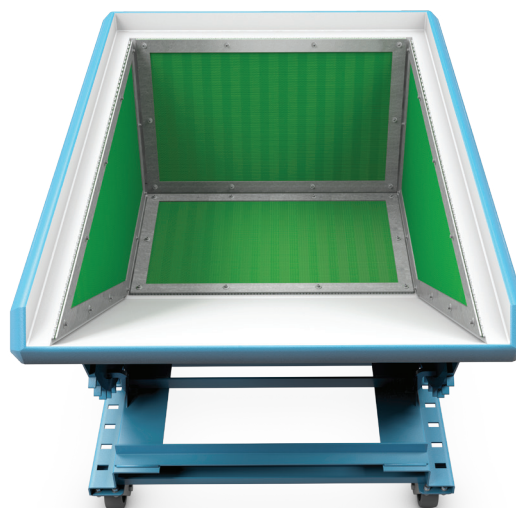
Log on to **ParkProcess.com** for more information.



Self Dumping Hopper

► Features

- Heavy duty carbon steel construction (optional stainless steel)
- Designed to be handled by a standard fork truck
- Typical (2) 2" drain ports located at the bottom on each side at opposing ends
- Filter media covers the floor and 4 walls
- Filter support panels made of perforated plates (optional materials available for support panels)
- Filter media comes in a choice of plastic or stainless steel and is held in place by bolts and brackets
- A double wall filter element that splits the hopper into two compartments is an available option
- Micron size filters from 2000 microns to 44 microns (10 mesh to 325 mesh)
- Dump Hopper sizes available: 1, 2, 3, 4, and 5 cubic yards



► Applications

- Waste Water Treatment Plants
- Vac Truck Waste
- Industrial Plant Sump Waste
- Truck & Heavy Equipment Wash Basin Waste

Dewatering Containers

- Self Dumping Hoppers
- Front Loaders
- Roll-Offs
- Permanently Mounted
- Tipping Stand
- Trailer Mounted
- Rotator Boxes
- Crane Dumping
- Filter Media

Filtration Equipment

- Cartridge Filter Vessels
- Cartridge Filters
- Bag Filter Vessels
- Bag Filters
- Filter Presses (Standard/Oilfield)
- Filter Press Cloths
- Sand Filters
- Back Wash Filters
- Carbon Filters
- Filter Media (Permanent /Disposable)

Separation Equipment

- Oil/Water Separators
- Hydrocyclones
- Vibratory Screening Units
- Inclined Plate Clarifier
- Grit Classifiers
- Strainers
- Oil Skimmers
- Mud Cleaners
- Centrifuges

Mixing/Blending

- Static Mixers
- Eductors
- Batch Mixing Systems
- Tank Mixers
- Agitators
- Blend Plants
- Blending Units
- Mud Shearing

Chemical Treatment

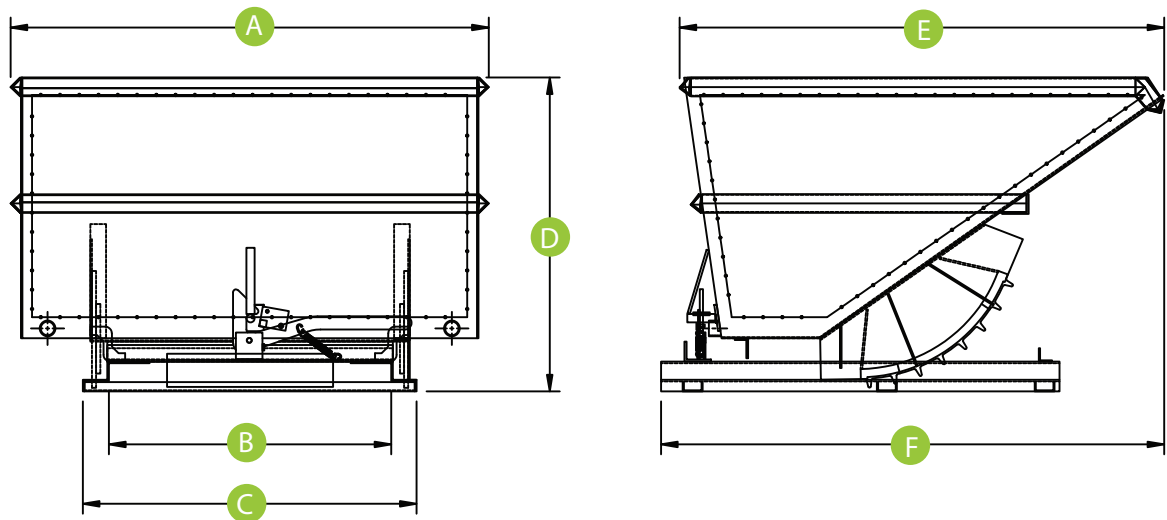
- Polymer Mixing & Injection
- Static Mixers
- PH Control Systems
- Slurry Tanks

Pumping

- Centrifugal Pumps
- Progressive Cavity Pumps
- Diaphragm Pumps
- Trash Pumps

Conveyance

- Screw Conveyors
- Drag Conveyors



Model	Size	A	B	C	D	E	F	Dry Weight
GCDH01	1 yd	39 1/2	30 3/4	38 3/8	37 3/4	62 1/2	64 3/4	660
GCDH02	2 yd	51 1/2	42 1/2	50 1/4	46 1/4	73	75 3/4	860
GCDH03	3 yd	72	42 1/2	50 1/4	47 1/4	73	75 3/4	1270
GCDH04	4 yd	99 1/2	42 1/2	50 1/4	47 1/4	73	75 3/4	1670
GCDH05	5 yd	123 1/2	42 1/2	50 1/4	47 1/4	73	75 3/4	1950

*Specifications subject to change without notice. Measurements in inches and weights in pounds.

Other available models:



Front Loader (FL)



Roll-Off (RO)



Signature Page

I accept this submittal package and release the equipment for fabrication.

Customer: Nan Inc / Promark

Job Number: N/A

Contact Name:

Signature: _____

Date: _____

Please sign this page and return to ParkProcess.